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Nonlinear Transient Thermo-elastic Analysis of a 2D-FGM Thick Hollow Finite Length Cylinder

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Abstract

In this Paper transient thermo-elastic analysis of a thick hollow finite length cylinder made of two dimensional functionally graded materials (2D- FGMs) has been investigated. The transient heat conduction and the thermo-elastic equations have been solved for the cylinder subjected to thermal loading utilizing finite element method. The higher order Lagrange shape function elements have been used to improve the accuracy of the temperature and thermal stress distribution in the axisymmetric thermo-elastic analysis. In addition, a new effective material model according to the Mori-Tanaka scheme for 2D- FGM has been developed and implemented. The nonlinear temperature-dependency of material properties as well as their variation along two directions led into highly nonlinear equations. In consequence, the temperatures and stresses have been evaluated in the different time frames and the effects of the material distributions in two radial and axial directions have been investigated. The results mainly indicate that the variation of the material distribution along the axial direction significantly influence the temperature and stress distributions. Also the material distribution along the z-direction increases the design variables, so it will be helpful to achieve more desirable results.

Keywords: 2D-FGM; temperature dependent properties; transient thermo-elastic; thick hollow cylinder; Mori-Tanaka material model.

Introduction

Functionally graded materials (FGMs) are composite materials that are formed of two or more constituent phases with a composition that is variable, continuously [1]. These materials at first have been proposed by Japanese scientists in the mid-1980s. FGMs have being used in industrial applications particularly in situations where the temperature variations are unavoidable, such as aerospace and fusion reactors.

Aboudi et. al. have presented the full generalization of the Cartesian coordinate-based higher-order theory for functionally graded materials. So they have illustrated both the fundamental issues related to the influence of microstructure on microscopic and macroscopic quantities governing the response of composites and the technologically important applications. A major issue addressed there was the applicability of the classical homogenization outlines in the analysis of functionally graded materials [2]. Ootao et al. have applied a neural network to an optimization problem of material compositions for a hollow circular cylinder of functionally graded material by aim of minimizing the thermal stress distribution. The thermal stress components have been formulated making use of the Airy's stress function method [3]. Birman and Byrd presented a review of principal developments in FGMs that includes heat transfer issues, stress, stability and dynamic analyses, testing, manufacturing and design, applications, and fracture [4]. They showed that most of early research studies in FGMs had more focused on thermal stress analysis and fracture mechanics. Zimmerman and Lutz studied the thermal stress and thermal expansion in a uniformly heated functionally graded cylinder [5].They proposed an analytical solution for solving the thermo-elasticity equations at the steady state. In order to solve the mathematical model, material properties were considered as exponential functions with continuous gradation.

Mechanical and thermal stresses in a functionally graded hollow cylinder due to radially symmetric load were studied by Jabbari et al. [6]. They also proposed a general solution for mechanical and thermal stresses in a functionally graded hollow cylinder due to non-axisymmetric steady state load [7].

Analysis of the thermal stress behaviour of functionally graded hollow circular cylinder was done by Liew and Kitipornchai [8]. They concluded that thermal stresses necessarily occur in the FGM cylinder, except in insignificant case of zero temperature. End effects of heat conduction in circular cylinders with finite length of functionally graded materials and laminated composites studied by Tarn and Wang [9]. They evaluated the decay length that characterizes the end effects on the thermal field. Chen and Tong studied thermo-mechanically coupled sensitivity analysis and design optimization of functionally graded circular cylinder under thermal and mechanical loading at the static and non- static state [10]. Mechanical and thermal stresses of a functionally graded circular hollow cylinder with finite length were studied by Shao [11]. The solutions of temperature, displacements, and thermal/mechanical stresses in a functionally graded circular hollow cylinder were considered in this research.

Analytical solution of 2D Non-axisymmetric elasticity and thermo-elasticity problems for radially inhomogeneous hollow cylinder have been proposed by Tokovoyy et.al. [12]. A solution of the equation for the governing stress in the form of Fourier series was presented. Hosseini kordkheili et al., have presented an analytical thermo-elasticity solution for hollow finite-length cylinders made of functionally graded materials exposed to thermal loads, internal pressure and axial loadings [13]. They concluded that stress and displacement components behave nonlinearly, especially for the regions near the longitudinal ends of the functionally graded cylinders. Nonlinear transient stress and wave propagation analyses of the FGM thick cylinders, employing a unified generalized thermo-elasticity theory have been investigated by Shariyat [14]. Nonlinear generalized and classical thermo-elasticity analyses were performed for functionally graded thick cylinders with temperature-dependent material properties subjected to various thermo-mechanical shocks at their inner and outer surfaces, employing Hermitian elements. There was also employed the Mori-Tanaka homogenization model of variations of the material properties instead of the simple rule of mixture.

Uniform surface heating of a functionally graded magneto-electro-thermoelastic hollow cylinder were analysed theoretically by Yoshihiro Ootaoet.al.[15]. They analysed transient thermal stress problem by using a laminated composite model and studied the effects of non-homogeneity of material on the stresses distribution. Haw-Long Lee et al., have investigated temperature distributions and thermal stresses in a functionally graded hollow cylinder under simultaneous inner and outer boundary heat flux and an inverse algorithm based on the conjugate gradient method and discrepancy principle applied in this study [16]. Hosseini et al., have investigated thermo-elastic waves propagation in functionally graded hollow cylinder based on Green-Naghdi model of coupled thermo-elasticity [17]. The governing equations are transferred to Laplace domain and solved analytically using series solution. Tariq Darabesh et al., have studied transient thermo-elasticity response of a FG hollow cylinder subjected thermal loading based on Green-Lindsay model [18]. The governing equations have been solved by using Galerkin finite element method. Temperature distribution in a functionally graded hollow cylinder under a transient-state temperature field based on thermal elasticity theory and the arbitrary difference precise integration (ADPI) has been investigated by Xiao-dan Zhang [19]. Thermo-elastic dynamic response of a long hollow FG cylinder subjected to radially symmetric dynamic mechanical and thermal loadings has been investigated numerically by Hong-Ling Dai et. al. [20]. The governing equations have been solved by finite difference and Newmark methods. Xie et al., have investigated 2-D thermo-elastic dynamic response of a long FG hollow cylinder under asymmetrical thermal and mechanical loadings [21]. The material properties are considered to be temperature independent. The governing equations have been solved by finite difference and Newmark methods. FG hollow cylinder's deformations and stress distributions affected by changing in volume fraction and temperature boundary conditions have been investigated. Inverse hyperbolic thermo-elastic problem of a functionally graded hollow cylinder has been solved by applying an inverse algorithm base on conjugate method and the discrepancy principle by Yu-ching Yang et al.

Nemat-Alla proposed the idea of adding a third material constitute to the conventional FGM constituents in order to withstand the serious thermal stresses [22]. The defined FGM has been considered as 2D-FGM. The study on 2D-FGM was carried out and pointed out that 2D-FGM has higher capability to relax thermal stresses than conventional FGM. In another study, Nemat-Alla et al. investigated the 2D-FGM plate under two-dimensional sever thermal loading with consideration of the temperature dependency of their properties and elastic–plastic behavior [23]. The obtained results in both cases were presented and compared in order to investigate and assess the effectiveness of 2D-FGM under such sever thermal loading conditions. Ching and Yen have investigated numerical solutions obtained by the meshless local Petrov-Galerkin (MLPG) method for 2D functionally graded solids, which is subjected to mechanical or thermal loads. In this MLPG analysis, the penalty method is used to efficiently apply the essential boundary conditions, and the test function is chosen to equal the weight function of the moving least squares approximation. Also two types of material gradations have been considered in which the first one has been based on the analytical functions (e.g. exponential or power law variation of material properties) and the other is based on the micromechanics model like Mori–Tanaka or self-consistent scheme [24].

Asgari and Akhlaghi have studied the effect of two-dimensional material distribution on the time response of the thick finite length cylinder structure and the thermal stress distributions were studied [25]. The finite element method with graded material properties within each element was used to model the material properties variations. Material properties were calculated by using linear rule of mixture and prescribed volume fraction in each point.

Shojaeefard and Najibi have investigated the nonlinear heat transfer of 2D-FGM hollow thick cylinder with finite length numerically. The effects of temperature dependency of materials has also been studied and concluded that for heat transfer and thermal stress analysis, the temperature dependency of materials properties should be considered for accurate results [26]. Molaei Najafabadi et al. have investigated the transient non-Fourier temperature and associated thermal stresses in a functionally graded slab symmetrically heated on both sides. Hyperbolic heat conduction equation in terms of heat flux has been utilized for obtaining temperature profile. So the separation of variables has been implemented to solve the non-homogenous thermoelastic problem. The material properties have been assumed to vary exponentially and they have indicated that the non-Fourier heat conduction has significant effect on the dynamic temperature and stress field [27]. Yang et al. [28] presented the new approach for 2D and 3D thermal stress analysis of FGMs by the aid of analytical expressions in radial integration boundary element method. Application of Kelvin's fundamental solutions leads to integral equations which includes both non-homogeneity and temperature variations of the material. Next, integrals have been transformed into equivalent boundary integrals using RIBEM. The authors have demonstrated efficiency of presented approach inserting: temperature, displacement and stress distributions of exemplary hexahedral structure. They also have concluded that the BEM-based method is competitive to conventional FEM-based methods since it allows saving more than 60 % of computational time. Ganczarski and Szubartowski demonstrated the plane stress state of FGM thick plate under thermal loading [29]. The Sneddon–Lockett theorem on the plane stress state in an isotropic infinite thick plate has been generalized for a case of 1D- FGM problem in which all thermo-mechanical properties were optional functions of depth coordinate. A nonlinear 1D thermo-elastic analysis of a thick-walled cylinder made of functionally graded material has been performed by Moosaie [30]. The temperature dependency of material properties on temperature has been considered. Moreover, the power law simple rule of mixture has been chosen for some thermal and mechanical properties. The perturbation technique has been employed to solve the nonlinear heat conduction equation and the temperature field has been then supplied to elasticity equations to extracting the thermal stresses [31]. Jabbari et al. have studied the stress and displacement distribution on the rotating truncated conical shells with varying thickness made of functionally graded materials (FGMs) subjected to thermo-mechanical loading. The materials have been assumed to be perfectly elastic and isotropic and vary according to the power law function in the axial direction of the conical shell. Based on steady-state heat conduction equation, the higher-order shear deformation theory (HSDT) has utilized and the system of partial differential equations is semi-analytically solved by using multi-layer method (MLM) [32].

It is obvious from the literature review that most of the researches on the thermo-mechanical behaviour of FGMs has been restricted to 1D- FGMs and very little work has been performed that considers the idea of adding more than two materials into structure; furthermore, the effective material models, which were used previously do not represent the actual behaviour of the FGM structures. The main objective of this paper is to analyse the thermo-mechanical behaviour of 2D- FGMs cylinder under abrupt thermal loading conditions. For this purpose, a new effective material model based on Mori-Tanaka scheme is developed for 2D-FGM and verified with previous experimental results. Then, the graded finite element method with higher order Lagrange elements are utilized and verified to analyse the thermo-mechanical problem. Finally, the temperature and the thermal stress distributions for different time frames and material distributions have been investigated. Specifically, parametric studies of the radial and axial material distribution compositions have been carried out.

Problem definition and governing equations

Volume fraction in 2D-FGM cylinder

The FGM cylinder in this study has the inner radius of $a = 0.1m$ and the outer radius $b = 0.15m$ and the length of $l = 0.2m$. The volume fractions of 2D-FGM at any point across the 2D-FGM axisymmetric cylinder wall are shown in Fig. (1). This scheme is estimated using the Eq. (1). The inner surface is made of two distinct ceramics and the outer surface consists of two different metals.

$$V_c = \left[1 - \left(\frac{r-a}{b-a}\right)^{n_r}\right] \tag{1a}$$

$$V_m = \left[\left(\frac{r-a}{b-a}\right)^{n_r}\right] \tag{1b}$$

$$V_1 = \left[1 - \left(\frac{r-a}{b-a}\right)^{n_r}\right]\left[1 - \left(\frac{z}{l}\right)^{n_z}\right] \tag{2a}$$

$$V_2 = \left[1 - \left(\frac{r-a}{b-a}\right)^{n_r}\right]\left[\left(\frac{z}{l}\right)^{n_z}\right] \tag{2b}$$

$$V_3 = \left[\left(\frac{r-a}{b-a}\right)^{n_r}\right]\left[1 - \left(\frac{z}{l}\right)^{n_z}\right] \tag{2c}$$

$$V_4 = \left[\left(\frac{r-a}{b-a}\right)^{n_r}\right]\left[\left(\frac{z}{l}\right)^{n_z}\right] \quad (2d)$$

where in Eqs. (1)-(2), subscripts 1, 2, 3 and 4 denote the first and second ceramic and the first and second metal respectively. Also n_r and n_z are parameters that represent the power exponent of volume fraction distributions in r and z-directions. The sum of the volume fractions of all the constituent materials is going to be 1. Therefore, it can be followed as:

$$\sum_{i=1} V_i = 1 \quad (3)$$

Homogenization method for Functionally Graded Materials

Generally, there are two approaches to homogenization of FGM. The choice of the approach should be based on the gradient of gradation relative to the size of a typical representative volume element (RVE) [4]. In the case where the variations of the material properties associated with gradation are relatively slow, standard homogenization methods can be applied. However, if the properties of the material vary rapidly with the coordinates, it is impossible to neglect the heterogeneous nature of RVE. In this case grading is reflected at both microscopic RVE as well as macroscopic structural scales; therefore the graded finite element in which the gradation of the material properties is interpolated through the element will be applicable here to consider the in-homogeneity in micro and macroscopic structural scale [33]. According to the works done in [34, 35], the Mori–Tanaka model which were developed for homogeneous particulate materials may yield satisfactory results in FGM subjected to uniform and non-uniform global loads.

Rules of mixture of FGM

As it is mentioned before, the Mori–Tanaka scheme for estimating the effective material properties is applicable to regions of the graded microstructure which have a well-defined continuous matrix and a discontinuous particulate phase. It is assumed that the matrix phase, denoted by the subscript 1, is reinforced by spherical particles of a particulate phase, denoted by the subscript 2 [36].

As presented in Reiter et al. and Reiter and Dvorak [34, 35], the Mori–Tanaka model has been shown to yield accurate prediction of the properties with a well-defined continuous matrix and discontinuous inclusions; therefore, the matrix and the inclusion will change by moving from the ceramic rich surface to metal rich surface. The axisymmetric cylinder region divided into 3 zones. First zone called as the ceramic zone, considers the ceramic as a matrix and the metal as an inclusion. The second zone called as the transition zone, in which skeletal microstructures characterized by a wide transition zone between the regions with prevalence of one of the constituent phases, is defined with an appropriate transition function. The third or metal rich zone considers the metal as a matrix and the ceramic as an inclusion.

Transition function

To accommodate the discontinuities in the homogenized material properties predicted at the boundaries of the different zones, the Eq. (7) defines the third order transition function. Let P_γ # P_β denotes the magnitudes of a certain effective material parameter predicted by two different averaging methods γ and β at a given volume fraction $V_i = V_{ib}$ of materials $c_i$and m_i as inclusion in a matrix $m_i$andc_i at the transition region. Moreover let [35]:

$$P = \mu\,(V_i)P_\gamma + (1 - \mu\,(V_i))P_\beta \quad (5)$$

Define the value of material parameter within a small interval $V_{ib} \pm \delta/2$, such that:

$$\mu\,(V_i) = 1 \qquad for \quad V_i < V_{ib} - \delta/2 \quad (6a)$$

$$\mu\,(V_i) = F_{trans.}(V_i) \qquad for \;\; V_{ib} - \delta/2 < V_i < V_{ib} + \delta/2 \tag{6b}$$

$$\mu\,(V_i) = 0 \qquad for \;\; V_i > V_{ib} + \delta/2 \tag{6c}$$

while the third-order transfer function was taken as:

$$F_{trans.}(V_i) = 2[(V_i - V_{ib})/\delta]^3 - 3(V_i - V_{ib})/(2\delta) + ½ \tag{7}$$

Rules of mixture of 2D-FGM

The rules of mixture for the 2D-FGM can be developed with some mathematical manipulation. For any point on the 2D-FGM axisymmetric cylinder region with volume fractions V_1, V_2, V_3 and V_4 as shown in Fig. (1), using Eqs. (4- 7), the rules of mixture for the 2D-FGM can be developed according to the below relations for K_f and the same mathematical manipulation for G_f, α_f and κ_f:

$$K_f = F_{trans.}(V_f) \times K_{fc} + (1 - F_{trans.}(V_f)) \times K_{fm} \tag{8a}$$

$$\frac{K_{fc} - K_c}{K_m - K_c} = \frac{V_m}{1 + (1 - V_m)(3(K_m - K_c)/(3K_c - 4G_c))} \tag{8b}$$

$$\frac{K_{fm} - K_m}{K_c - K_m} = \frac{V_c}{1 + (1 - V_c)(3(K_c - K_m)/(3Km - 4G_m))} \tag{8c}$$

$$\frac{K_{fm} - K_m}{K_c - K_m} = \frac{V_c}{1 + (1 - V_c)(3(K_c - K_m)/(3Km - 4G_m))} \tag{8c}$$

$$K_c = F_{trans.}(V_c) \times K_{c1} + (1 - F_{trans.}(V_c)) \times K_{c2} \tag{8d}$$

$$K_{c1} = \frac{V_2(K_2 - K_1)}{1 + (1 - V_2)(3(K_2 - K_1)/(3K_1 - 4G_1))} + K_1 \tag{8e}$$

$$K_{c2} = \frac{V_1(K_1 - K_2)}{1 + (1 - V_1)(3(K_1 - K_2)/(3K_2 - 4G_2))} + K_2 \tag{8f}$$

$$K_m = F_{trans.}(V_m) \times K_{m1} + (1 - F_{trans.}(V_m)) \times K_{m2} \tag{8g}$$

$$K_{m1} = \frac{V_4(K_4 - K_3)}{1 + (1 - V_4)(3(K_4 - K_3)/(3K_3 - 4G_3))} + K_3 \qquad (8h)$$

$$K_{m2} = \frac{V_3(K_3 - K_4)}{1 + (1 - V_3)(3(K_3 - K_4)/(3K_4 - 4G_4))} + K_4 \qquad (8i)$$

The procedure applied for obtaining bulk module K_f will be applicable for the G_f, α_f and κ_f. The ρ_f and C_f are calculated according the below equations for both of the models [23]:

$$\rho_f = \rho_1 V_1 + \rho_2 V_2 + \rho_3 V_3 + \rho_4 V_4 \qquad (9)$$

$$C_f = \frac{C_1\rho_1 V_1 + C_2\rho_2 V_2 + C_3\rho_3 V_3 + C_4\rho_4 V_4}{\rho_1 V_1 + \rho_2 V_2 + \rho_3 V_3 + \rho_4 V_4} \qquad (10)$$

where,"f" index shows the effective material property in Eqs. (8)-(10).As FGMs mainly are used in the thermal barrier components, it is important to take into account the temperature dependency of material properties during the thermal stress analysis. The properties of materials are assumed to be temperature dependent and can be expressed as a nonlinear function of temperature [1]:

$$P_i = P_0(P_{-1}T^{-1} + 1 + P_1 T + P_2 T^2 + P_3 T^3) \qquad (11)$$

where P_i is the specific material properties for each constituents and P_0, P_{-1}, P_1, P_2 and P_3 are the coefficients of temperature T (in K) and are unique to the constituent materials. Here, the density of the material is a function of temperature and coefficient of thermal expansion as [23]:

$$\rho_i = \frac{\rho_0}{[1 + \alpha_i(T)]^3} \qquad (12)$$

Typical values for different thermal and mechanical properties of ceramics and metals are listed in reference [1].

Thermo-mechanical properties validation

To display the capability of the present model to predict the thermo-mechanical material properties, the modeling predictions have been compared with the experimental data in which the coefficient of thermal expansion and Young's modulus of the Al/Al_2O_3 composites at room temperature have been determined.

According to Neubrand and et al. [37], the Al/Al2O3 FGMs have been prepared by an adaptation of the gradient materials by foam compaction (GMFC) replication process. The samples are plates with geometry of (35 × 33 × 3 mm3) and a non-homogeneous Al content along the longest axis. Half of the plate is always consisted of Al/Al_2O_3 composite with an aluminum volume fraction of 2.5%. The other half of the plate (a region with length of 17.5 mm) contains a composition gradient. The material properties of Al and Al_2O_3 have been specified in table 1 [37].

Fig. (2) demonstrates the comparison of the effective coefficient of thermal expansion proposed in this study (M-T) with experimental results. The proposed model was found to fit reasonably well with their experimental results, which the maximum error is about 4.5%. Accordingly, comparison of Young's modulus determined by proposed model (M-T) and experimental results has been shown in Fig. (3), which indicated that the proposed model was found to fit well with the experimental results.

The Young's modulus contour for $n_r = 2$ and $n_z = 1$ at time frame $t = 300$ seconds has been illustrated in Fig. (4). This contour has been derived from the new material model based on Mori-Tanaka scheme for 2D-FGM.

Subsequently, Fig. (5) shows that heat conductivity coefficient contour for $n_r = 2$ and $n_z = 1$ at $t = 200$ seconds,in which the lowest value of heat conductivity coefficient has been related to ZrO_2 (Ceramic 1).

Graded finite element modelling

Graded elements can be compared with conventional homogeneous elements such as those used in traditional layered analysis of FGMs. It would be noted that the graded element incorporates the material property gradient at the size scale of the element, while the homogeneous element produces a stepwise constant approximation to a continuous material property field. The special effects of these discontinuities will be more considerable in the 2D-FGMs because of its 2D non-homogeneity. Graded elements are implemented by means of direct sampling properties at the Gauss points of the element [33, 38]. Based on these facts the graded finite element is strongly desirable for modelling the present problem.

Heat transfer equations

The equation of heat conduction in the axisymmetric cylindrical coordinate without heat sources is demonstrated as [39]:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(rk(r.z.T)\frac{\partial T(r.z.t)}{\partial r}\right)+\frac{\partial}{\partial z}\left(k(r.z.T)\frac{\partial T(r.z.t)}{\partial z}\right) = \rho(r.z.T)c(r.z.T)\frac{\partial T(r.z.t)}{\partial t} \quad (13)$$

where $k(r.z.T)$, $\rho(r.z.T)$ and $C(r.z.T)$ are thermal conductivity, mass densityand heat capacity of functionally graded material, respectively. The thermal conductivity tensor is diagonal and $k_{rr}(r.z.T)$=$k_{zz}(r.z.T)$=$k(r.z.T)$.The thermal boundary conditions can be formulated as:

$$-k\frac{\partial T(r=a.z.t)}{\partial r}+h_i(T-T_\infty)=0 \tag{14a}$$

$$T(r=b.z.t)=T_0 \tag{14b}$$

$$-k\frac{\partial T(r.z=0.t)}{\partial z}+h_L(T-T_\infty)=0 \tag{14c}$$

$$\mathrm{k}\frac{\partial T(r.z=L.t)}{\partial z}=0 \tag{14d}$$

where in Eq. (14) h_i and $h_L$are convection coefficients in the inner and lower surfaces, respectively and $T_\infty$is the surrounding temperature. The uniform initial temperature of the cylinder is $T_{initial}=327$°C; besides, the other specifications are denoted as: $h_i=h_L=750\ \frac{W}{m^2K}$;$T_\infty=27$°C and $T_0=90$ °C.

For modelling the heat transfer problem an axisymmetric ring element with rectangular cross-section is used to discrete the domain. Taking into account the nine nodal values oftemperature defined as the degrees of freedom, the quadratic interpolation model can be assumed as [39]:

$$T^e(r.z.t)=[N_i(r.z)]\{q_i(t)\}^e\ .\ i=1.2.\dots.9 \tag{15}$$

where $[N_i(r.z)]$ is the matrix of quadratic interpolationfunctions and$\{q(t)\}^e$ is the nodal temperature vector of nine nodal elements.The weak form of the present problem leads to the following integral:

$$\iiint_{\Omega}\left[k\big(r.z.T(r.z)\big)\left(\frac{\partial T}{\partial r}\right)^2+k\big(r.z.T(r.z)\big)\left(\frac{\partial T}{\partial z}\right)^2+2\rho cT\left(\frac{\partial T}{\partial t}\right)\right]d\Omega$$
$$+\iint_{\Gamma}h(T^2-2TT_{\infty})d\Gamma=0 \tag{16}$$

where Ω and Γ are the volume of the cylinder and the boundary conditions, respectively. The coefficient of thermal conductivity, heat capacity and density depend on $r.z$ and$T$. Considering the initial and boundary conditions the graded finite element equation for whole domain was derived as:

$$[K_M]\{\dot{Q}\}\ +\ [K]\{Q\}\ =\ F_T \tag{17}$$

where $\{Q\}$is the vector of nodal temperature of the cylinder and also:

$$[K_M]=\sum_{e=1}^{m}[k_m]^e \tag{18a}$$

$$[K]=\sum_{e=1}^{m}([k_1]^e+\ [k_2]^e) \tag{18b}$$

$$F_T=\{f_T\}^e \tag{18c}$$

The axisymmetric finite element model, used in the transient heat conduction study, contains 3,200 nine-node quadrilateral elements. This numbers of elements results from uniform dividing of the cylinder wall into 80 elements through the axial direction and 40 elements through the radial directions. These are the minimum number of elements concluded from mesh dependency analysis.

The global matrix equation for the structure can be obtained with assembling the element matrices such a one presented in Eq. (17). This set of equations is time dependent and highly nonlinear. Therefore, the method of the solution for such these large sets of equations needs the procedure in which the time dependence is given by a finite difference approximation [39]. Thus, the nonlinear solver uses an affine invariant form of the damped Newton method as described in [40].

Temperature gradient and nodal temperature evaluated from the numerical solution of the heat transfer equation will be utilized in the equilibrium and strain displacement equations to extract the thermal stresses as demonstrated in the following section.

Thermo-elasticity equations

For evaluation of thermal stresses due to temperature gradient the classical thermo-elasticity is used. Therefore, the equations of equilibrium for axisymmetric cylinder coordinates without body force have been considered

The linear strain–displacement relations have been utilized in this study:

$$\varepsilon_{rr} = \frac{\partial u}{\partial r}.\ \varepsilon_{\theta\theta} = \frac{u}{r}.\ \ \varepsilon_{zz} = \frac{\partial w}{\partial z}.\ \ \varepsilon_{rz} = \frac{1}{2}\left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial r}\right) \tag{19}$$

where, u and w are the displacement components along the r and z directions respectively.

The upper and lower surfaces of cylinder are roller constrained to prevent from axial displacement. Additionally, the inner surface of cylinder is fully constrained while the outer surface is free.

$$u(a.z) = w(a.z) = w(r.l) = w(r.0) = 0 \tag{20a}$$

$$\sigma_{rr}(b.z) = \ \sigma_{rz}(b.z) = 0 \tag{20b}$$

In order to model the elastic problem an axisymmetric ring element with rectangular cross-section is used to discrete the domain. Third order interpolation with

Lagrange shape function (cubic) that each element has sixteen nodes considered for demonstrating the stress gradients within the cylinder wall correctly.

The axisymmetric finite element model, used in the current study of the thermo-elastic problem, contains 4,800 sixteen-node quadrilateral elements. These numbers of elements results from uniform dividing of the cylinder wall into 120 elements through the axial direction and 40 elements through the radial directions. These are the minimum numbers of elements obtained from mesh dependency analysis.

The numerical solutions of the thermo-elastic investigation are conducted using the parallel sparse direct linear solver MUMPS (MUlti-frontal Massively Parallel sparse direct Solver) that works on general systems of form Ax = b. Finally, the stresses on each node were averaged according to the stresses on its associated elements and shape functions.

Finite Element Verification Example

To verify the present study, since similar works are few, the present problem is specified into a special case to compare with the ones available in literature. In this example results are derived for the very special case of the classical thermo-elasticity where the transient thermal stress distribution is considered. An axisymmetric, thermo-elastic boundary value problem is solved for an infinitely long functionally gradient cylinder with an inner radius of $0.0127m$ and an outer radius of $0.0254m$. The effects of the temperature-dependency of the material properties on the results are also considered. In this regard, a Si_3N_4/SUS304 cylinder with a ceramic-rich inner layer and a metal-rich outer layer is considered. The volume fraction distribution is chosen in terms of a power-law (n). The temperature-dependent material properties are presented in the reference [1].

All points of the cylinder are initially assumed to be at the ambient temperature of $298.15K$. The inner surface of the cylinder is suddenly subjected to a convection heat transfer with a transient medium with $2000K$ temperature and $h_{conv.} = 750\ \frac{W}{m^2 K}$. . Therefore, the boundary conditions are those of convective heat transfer and fixed radial displacement at the inner radius and insulated boundary and stress free condition set at the outer surface [1]:

$$2\pi r\left[k\frac{\partial T}{\partial r} + h_{conv.}(T - T_\infty)\right] = 0 \qquad u_r = 0 \qquad at\ \ r = r_i \tag{21a}$$

$$2\pi r k\frac{\partial T}{\partial r} = 0 \qquad \sigma_{rr} = 0 \qquad at\ \ r = r_o \tag{21b}$$

The hoop stress distribution at the inner surface of the cylinder as a function of time for two power exponents (n =1.0, and 25.0) are shown in Fig. (6).In this figure, the results from the present study are compared with the Reddy and Chin [1]. It is well observed that an excellent agreement is occurred comparing the results. The maximum error is estimated for $n = 25$ which is about 4.5 %. The higher order and graded finite elements formulation that have been applied for this study leads to more accurate results in comparison with the works done by Reddy and Chin.

Results and Discussion

The yield stress across the cylinder wall can be obtained from below equation:

$$\sigma_{yf} = \sigma_{y1}V_1 + \sigma_{y2}V_2 + \sigma_{y3}V_3 + \sigma_{y4}V_4 \tag{22}$$

where σ_{yf} is the effective yield stress and σ_{yi} are the yields stresses for each material respectively [23]. The effective stress can be evaluated from the principal stresses using the distortion energy theory or von Mises-Hencky theory and can be compared with the yield strength of the material for failure [1].The effective stress (von-Mises) is defined applying the σ_1, σ_2 and σ_3 as the principal stresses.

$$\sigma_{eff} = \left\{\frac{1}{2}\left[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2\right]\right\}^{1/2} \tag{23}$$

Temperature distribution through the cylinder wall after 300 seconds has been depicted in Fig. (7). The maximum value of temperature which is measured for $nr = 1$ and $nz = 5$ is about 142 °C.

Fig. (8) illustrates radial displacement distribution through the cylinder wall for $n_r = n_z = 2\ at\ t = 300$ seconds. The maximum value of radial distribution is about 0.185 mm on the outer surface of cylinder wall. Subsequently, axial displacement contour of the cylinder wall for the same time and material distribution has been demonstrated in Fig. (9). The maximum value of axial displacement is about 0.375 mm on the outer surface of the cylinder wall.

The radial and axial stresses as well as shear and hoop stresses for $nr = nz = 2\ at\ t = 300$ seconds have been demonstrated in Figs. (10) and (11) respectively. The higher values of stresses have been observed for axial and hoop stresses through the cylinder wall, while the radial and shear stresses illustrate lower ones.

The contours of temperature distributions for two different time frames have been depicted in Fig. (12), whereas the material distributions have been determined by $n_r = 1\ and\ n_z = 5$. It is obvious that the more the time passes, the more the temperature has been decreased in which the maximum value of temperature for $t = 150$ seconds is about 30% more than the temperature measured for $t = 250$ seconds.

Fig. (13) shows the contour of effective stress through the cylinder wall for $n_r = n_z = 2$ and different time frames of 200 and 300 seconds. Since the temperature decreases by passing the time, the value of effective stress has been increased. The maximum value of effective stress for $t = 200$ seconds is 47% more than the maximum one for $t = 300$ seconds.

The effect of various time frames have been investigated for effective young's modulus along the horizontal center line (HCL) versus normalized radial direction at $n_r = n_z = 2$. Fig. (14) demonstrates that the values of effective young's modulus have been increased over time. It is due to the fact that; the temperature has been decreased by convective cooling. It is obvious that the higher value of temperature leads to reduction of the effective young's modulus through the cylinder wall.

The values of the hoop stresses according to the metallic volume fraction (Vm) along the HCL at $n_r = n_z = 2$, for different time frames have been demonstrated in Fig. (15).These values have been increased for $Vm < 0.1$ and then take the maximum values for all the time frames; whereas, the highest value of the hoop stress has been devoted to $t = 100$ seconds. The high gradient of hoop stresses because of severe temperature gradient has been demonstrated for ceramic reach part of cylinder wall.

The ratio of effective stress versus effective yield stress called as normalized effective stress (NES) has been investigated in Fig. (16). NES along the HCL for $n_r = n_z = 2$ and different time frames has been illustrated. It can be concluded from this figure that the values of NES have been decreased over the time. It is due to the fact that, the effective yield stress has been increased by reduction of the temperature during the convection cooling, while, on the contrary the effective stress behaves in opposite way. By passing the time due to decreasing the temperature, the thermal stresses decrease and then the value of the NES is getting smoother.

Fig. (17), illustrates the variation of temperature versus V_m along the HCL for $n_z = 2$ and different values of n_r at t=300 seconds. It is well observed here, with increasing the value of n_r, for $V_m's$ more than 0.6, the temperature has been significantly increased in such a way that at $Vm = 0.9$ the highest value of temperature for $n_r = 5$ is 60% more than the one for $n_r = 0.1$. For $V_m's$ less than 0.6 the temperature follow steady pattern, due to the fact that the ceramic rich cylinder wall is intended to be cooled later because of its lower thermal diffusivity coefficient.

The temperature variation along HCL at $n_r = 2$ and different values of n_z at t=300 seconds have been illustrated in Fig. (18). On the contrary, these curves get their maximum values in the higher values of n_z. In addition, with increasing the value of n_z, temperature has been increased, significantly. The results show that the difference between the highest and lowest value of the temperature is about 60%. Moreover, the temperature is decreased in higher values of metallic volume fraction. The cylinder wall made of ZrO_2/SUS304 intends to be cooled sooner with a low temperature gradient as a result of its higher value of thermal diffusivity coefficient.

Fig. (19) depicts radial displacement versus Vm along the HCL for $n_z = 1$ and different values of n_r. It is obvious that the lower values of nr lead into lower radial displacement especially for higher volume fraction of metallic phase. No significant discrepancy is occurred for different nr in lower metallic volume fraction.

Radial displacement versus Vm along the HCL for $n_r = 1$ and different values of n_z has been illustrated in Fig. (20). As depicted in this figure, with increasing the value of n_z the radial displacement has increased in such a way that for $Vm = 1$, the highest value of radial displacement is more than twice of the lowest value of radial displacement. In other hand, with increasing the metallic volume fraction, the radial displacement curves are following ascending pattern.

To study the effect of radial and axial power exponents on normalized effective stress at centre point (CP) of cylinder wall, Figs. (21 and 22) demonstrate NES versus time at CP for different values of n_r and n_z, respectively. As depicted in Fig. (21), the values of NES have been decreased over time. The difference between NES curves is higher in the beginning of cooling process in which the temperature is higher. It is also interesting that the results at $n_r = 5$ depict lower values of NES than those of $n_r = 2$. However, at the end of the process two curves indicate the similar results.

The variation of axial power exponent has significant effect on NES. It is well demonstrated in Fig. (22) that with increasing the nz, the values of NES have been decreased significantly. The maximum value of NES for $n_z = 0.1$ is about three times more than this value for $n_r = 5$. During the cooling process the value of NES is following descending pattern in which after 200 seconds the NES curves are demonstrating the steady pattern.

As it is mentioned previously, the power exponent for radial and axial direction volume fraction variation (n_r and n_z) determine the variation of material properties through the cylinder wall. Now, the effects of $n_r \, and \, n_z$ on temperature and normalized effective stress on centre point of cylinder wall at the special time frame of 300 seconds have been investigated in Figs. (23, 24). For specific value of n_z except $n_z = 0.1$, with increasing the value of $nz$the temperature values have been increased. Also the difference between temperature values have increased by increasing the value of n_z, while for $n_z = 0.1$, it behaves in opposite way (Fig. 23). It means that the metallic rich cylinder wall becomes cool sooner because of higher coefficient of thermal diffusivity.

The lowest value of temperature is devoted to $n_r = 5$ and $n_z = 0.1$; furthermore, $n_r = n_z = 5$ has depicted the highest value of temperature at CP. In other words, the ceramic rich cylinder wall in which ceramic 2 dominates the region will have the lowest value of temperature at this point after 300 seconds.

Normalized effective stresses at centre point of cylinder wall after 300 seconds versus different values of n_r and n_z have been demonstrated in Fig. (24). For a specific value of n_z except $n_z = 0.1$, by increasing of n_r, NES has been increased, but for $n_z = 0.1$ this value has been decreased. Furthermore, except for $n_r = 0.1$ the variation of n_r and n_z does not have significant change in the values of NES. It is obvious that the lowest value of NES is devoted to $n_r = n_z = 0.1$ and the highest value of NES has devoted to $n_r = 0.1$ and $n_z = 5$. It means that whereas the Ti-4V-6Al is dominant through the cylinder wall, it will be able to withstand higher thermal stresses while the cylinder made up of SUS304 is suffers from thermal leakages. Moreover, for $n_z = 1$ the variation of n_r has not any effect on the value of NES at CP of the cylinder wall at 300 seconds. It has been also demonstrated that for $n_z = 1$, the changing of n_r will not affect the NES value. In other word, the material properties variation along the z-direction has been shown lower effective stresses. Therefore, it will be desirable for designer to outline thermal barrier cylindrical components.

Conclusion

The nonlinear transient heat conduction and thermo-elastic analysis have been conducted for the 2D- FGM cylinder subjected to thermal loading utilizing graded finite element method. Accordingly, a new effective material model according to the Mori-Tanaka scheme for 2D- FGM has been developed and implemented. Particularly, some major achievements are followed as:

1- Implementation of the Mori- Tanaka and third order transition function for 2D- FGM and validation of the process.

2- In order to achieve the more precise results, the non-linear temperature dependency of the material properties is taking into account. The results have been also validated with those of 1D-FGM model.

3- The temperatures and stresses at different time frames have been investigated. In addition, the effects of material distributions in two radial and axial directions have been considered.

4- It has been also indicated that the variation of the material distribution along the axial direction significantly influence the temperature and stress distributions.

5- By introducing the NES (Normalized Effective Stress) parameter, the cylinder wall strength has been studied for different values of n_r and n_z.

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Figure List:

Figure 3: Young's modulus as a function of Al volume fraction for Al/Al_2O_3 composites.

Figure 4: Young's modulus contour for $n_r = 2$ and $n_z = 1$ at time frame $t = 300\ s$.

Figure 5: Heat conductivity coefficient contour for $n_r = 2$ and $n_z = 1$ at time frame $t = 200\ s$.

Figure 6: The comparison of hoop stress distribution as a function of time.

Figure 7: Temperature distribution contour through the cylinder wall for $n_r = 1\ and\ n_z = 5$ after 300 seconds.

Figure 8: The contour of the radial displacement distribution through the cylinder wall for $n_r = n_z = 2\ at\ t = 300$ seconds.

Figure 9: The contour of the axial displacement distribution through the cylinder wall for $n_r = n_z = 2\ at\ t = 300$ seconds.

Figure 10: The radial and axial stresses distributions for $n_r = n_z = 2\ at\ t = 300$ seconds.

Figure 11: The shear and hoop stresses distributions for $n_r = n_z = 2\ at\ t = 300$ seconds.

Figure 12: The contours of temperature distributions for two different times (t=150 and 250 s) for $n_r = 1\ and\ n_z = 5$.

Figure 13: The contours of effective stresses through the cylinder wall for $n_r = n_z = 2$ and different time frames at 200 and 300 seconds.

Figure 14: Effective Young's modulus along the horizontal centreline HCL versus normalized radial direction for $n_r = n_z = 2$ at different time frames.

Figure 15: Transient hoop stresses according to the metallic volume fraction (Vm) along the HCL for $n_r = n_z = 2$and different time frames.

Figure 16: Transient normalized effective stress (NES) versus V_m along the HCL for $n_r = n_z = 2$ and different time frames.

Figure 17: Temperature versus Vm along the HCL for $n_z = 2$ and different values of n_r at $t = 300$ seconds.

Figure 18: Temperature along HCL versus Vm for $n_r = 2$ and different values of n_z at $t = 300$ seconds.

Figure 19: Radial displacement versus Vm along the HCL for $n_z = 1$ and different values of n_r at $t = 300\ s$.

Figure 20: Radial displacement versus Vm along the HCL for $n_r = 1$ and different values of $n_z\ at\ t = 300\ s$.

Figure 21: NES versus time at CP for different values of n_r and $n_z = 1$.

Figure 22: NES versus time at CP for different values of $n_z\ and\ n_r = 1$.

Figure 23: Temperature values at CP for different values of n_r and n_z.

Figure 24: NES values at CP for different values of n_r and n_z.

Table list:

Table 1: The material properties of Al and Al_2O_3 [37].

Table 1: The material properties of Al and Al_2O_3 [37].

	E (GPa)	??	α (1/°C)
Al	69	0.33	23.1×10^{-6}
Al_2O_3	390	0.2	7.7×10^{-6}

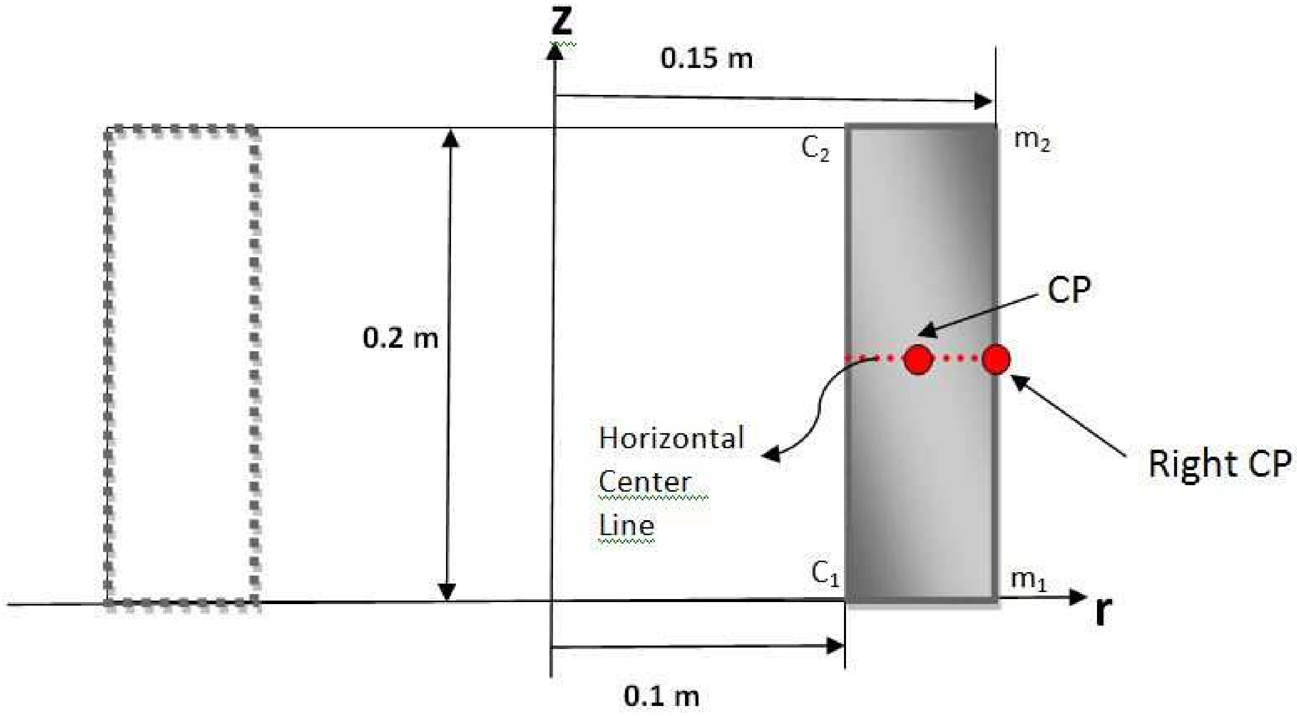

Figure 1: 2D-FGM and material distribution of the axisymmetric cylinder with predetermined points and line (C_1=ZrO_2, C_2=Si_3N_4 / m_1= SUS304, m_2= Ti-6Al-4V).

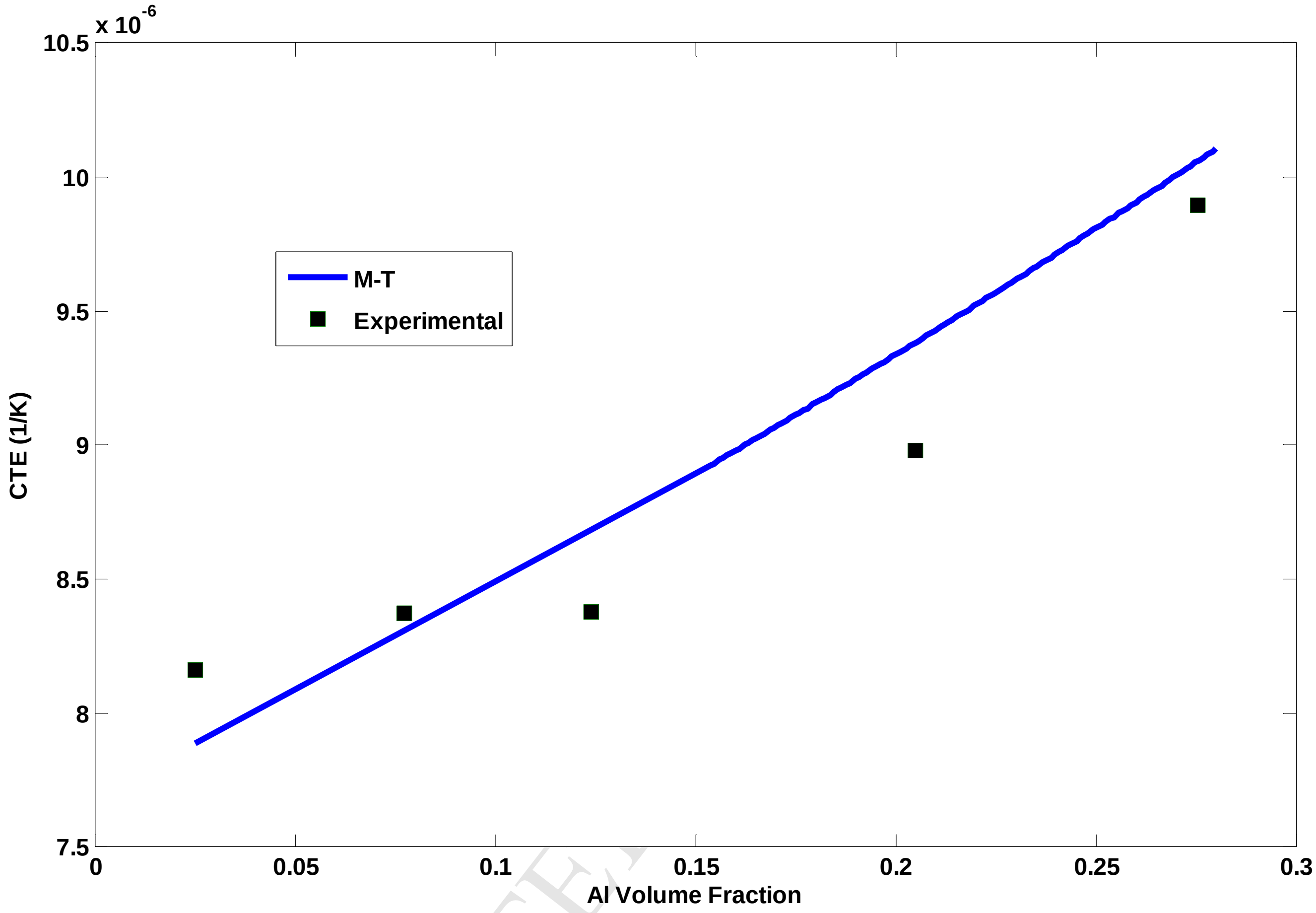

Figure 2. Comparison of the effective CTE distribution in Al/Al_2O_3 FGM.

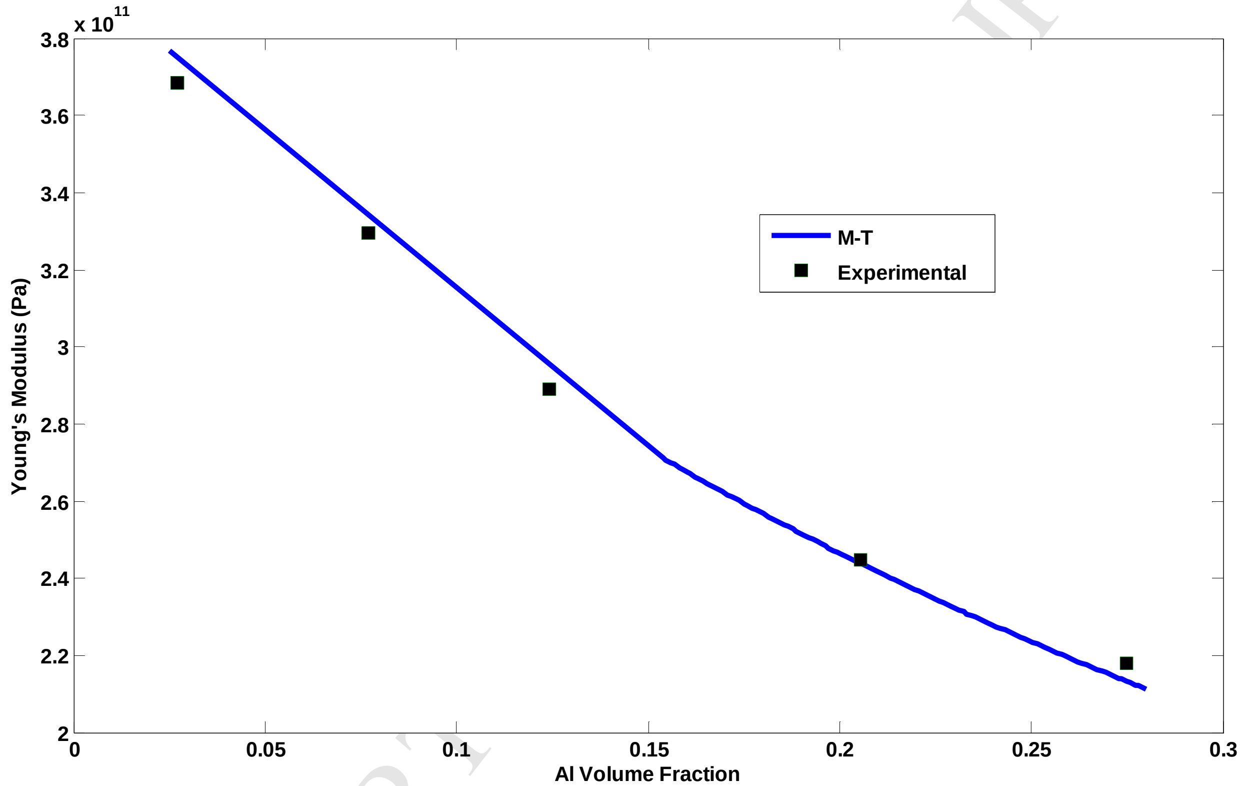

Figure 3. Young's modulus as a function of Al volume fraction for Al/Al_2O_3 composites.

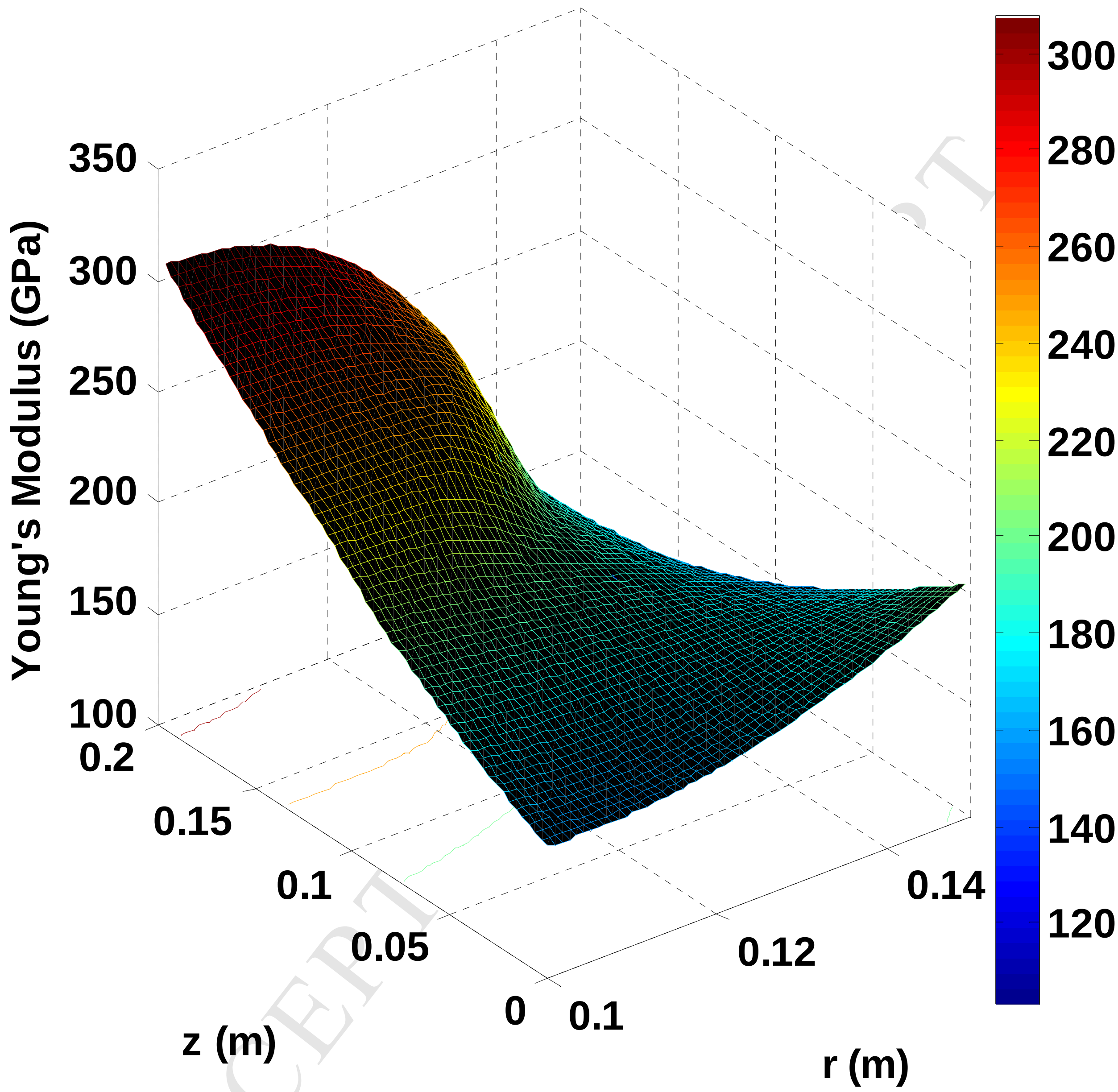

Figure 4: Young's modulus contour for $n_r = 2$ and $n_z = 1$ at time frame $t = 300\ s$.

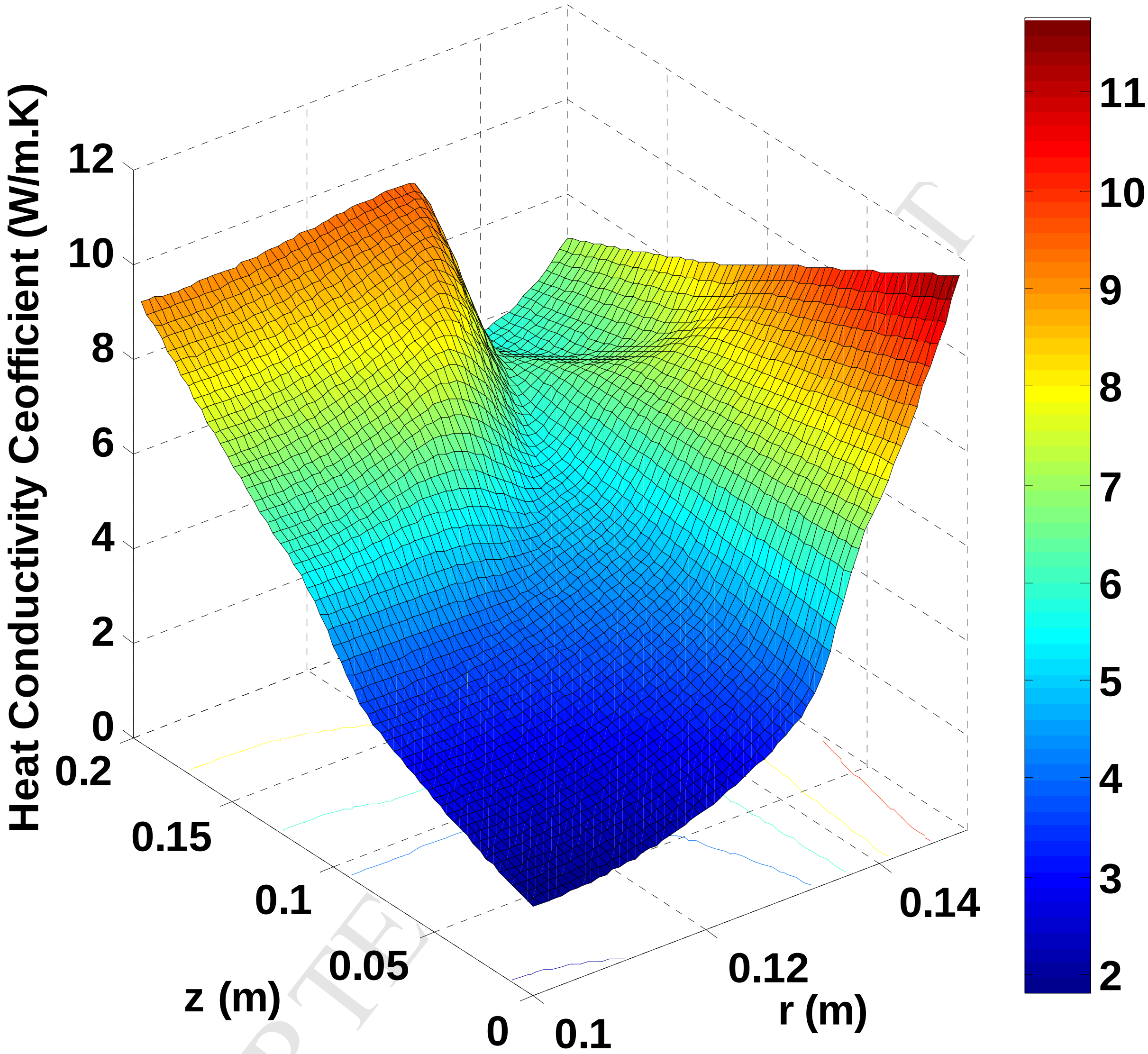

Figure 5: Heat conductivity coefficient contour for $n_r = 2$ and $n_z = 1$ at time frame $t = 200\ s$.

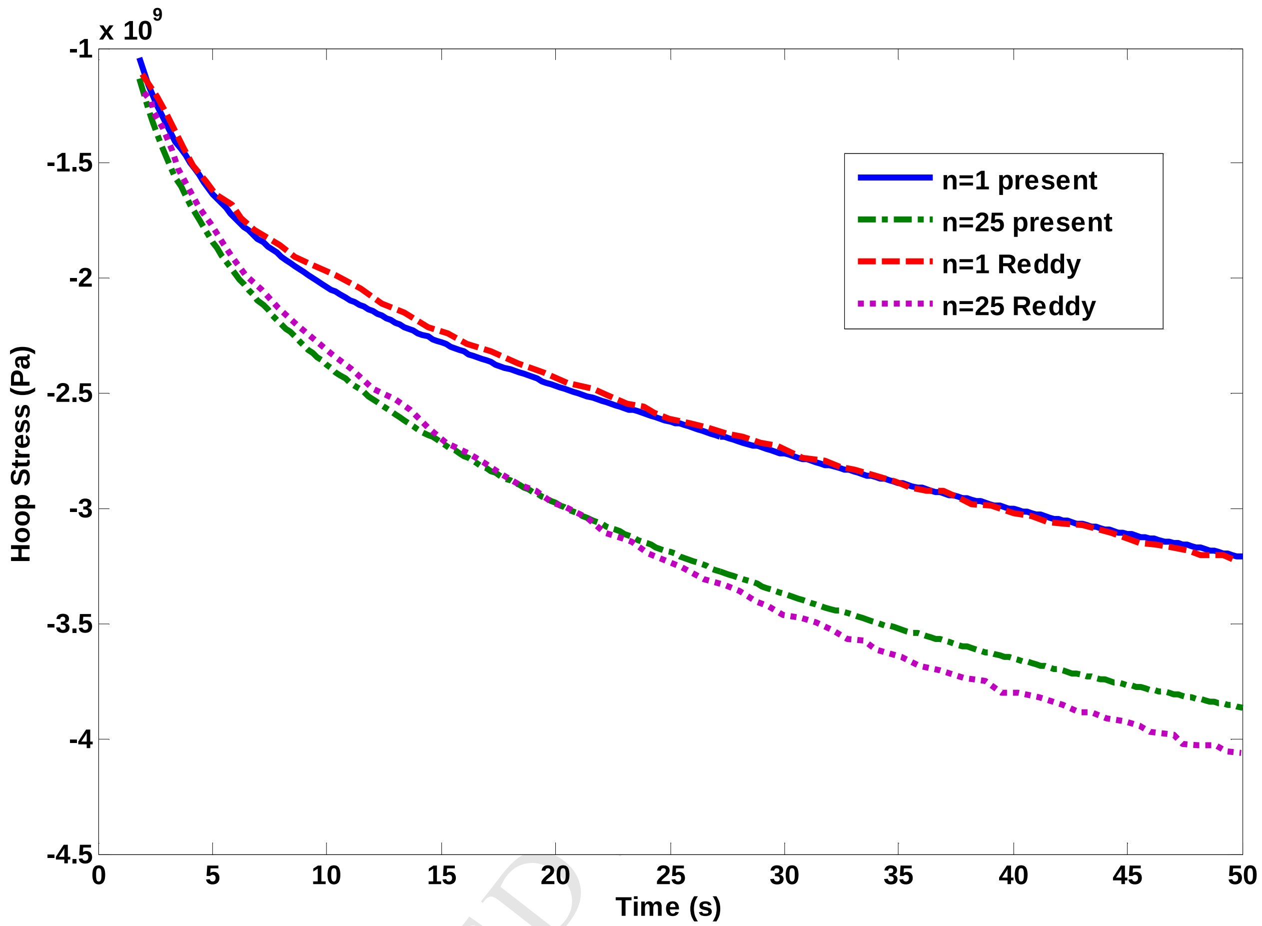

Figure 6: The comparison of hoop stress distribution as a function of time.

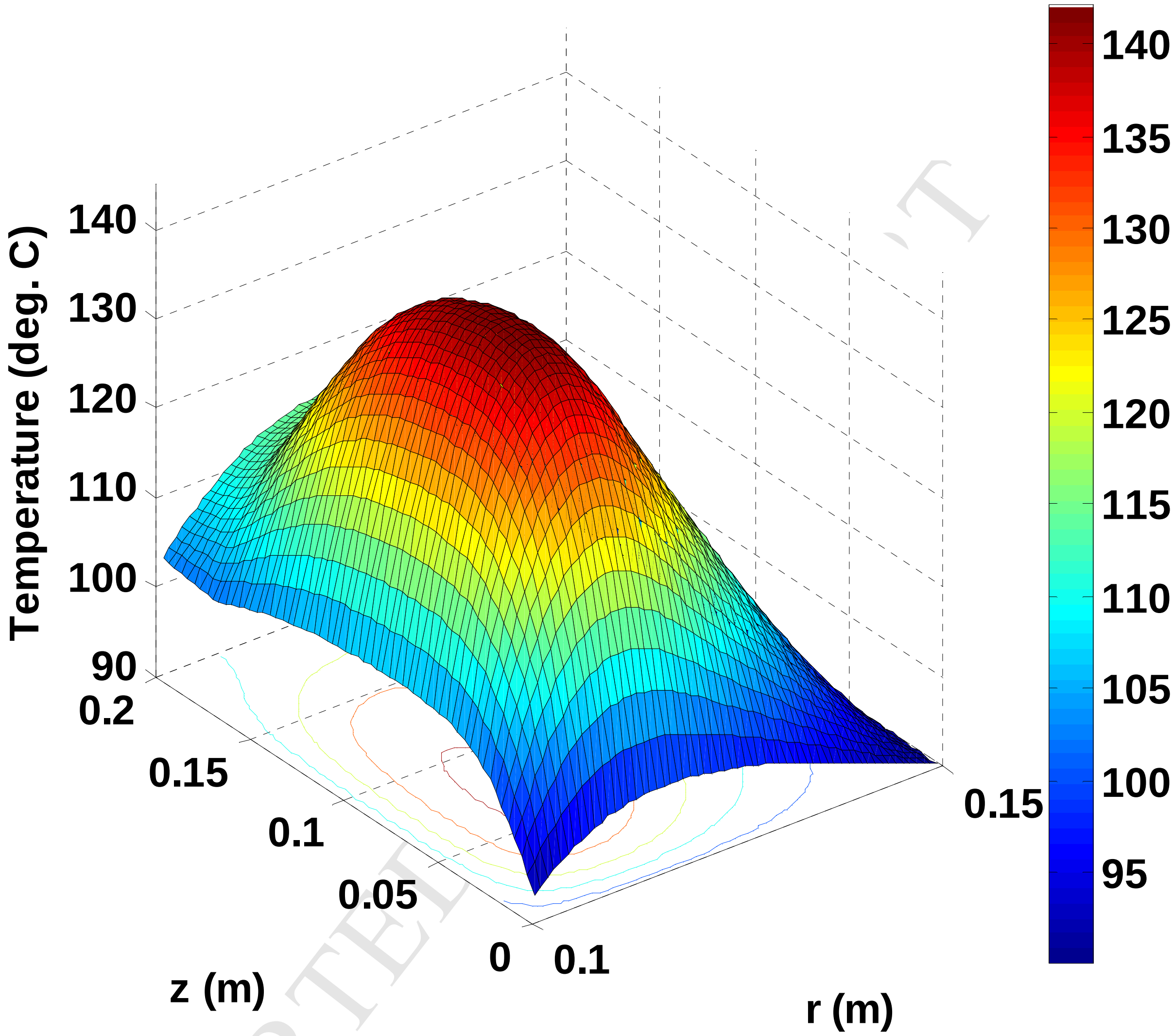

Figure 7: Temperature distribution contour through the cylinder wall for

$n_r = 1$ and $n_z = 5$ after 300 seconds.

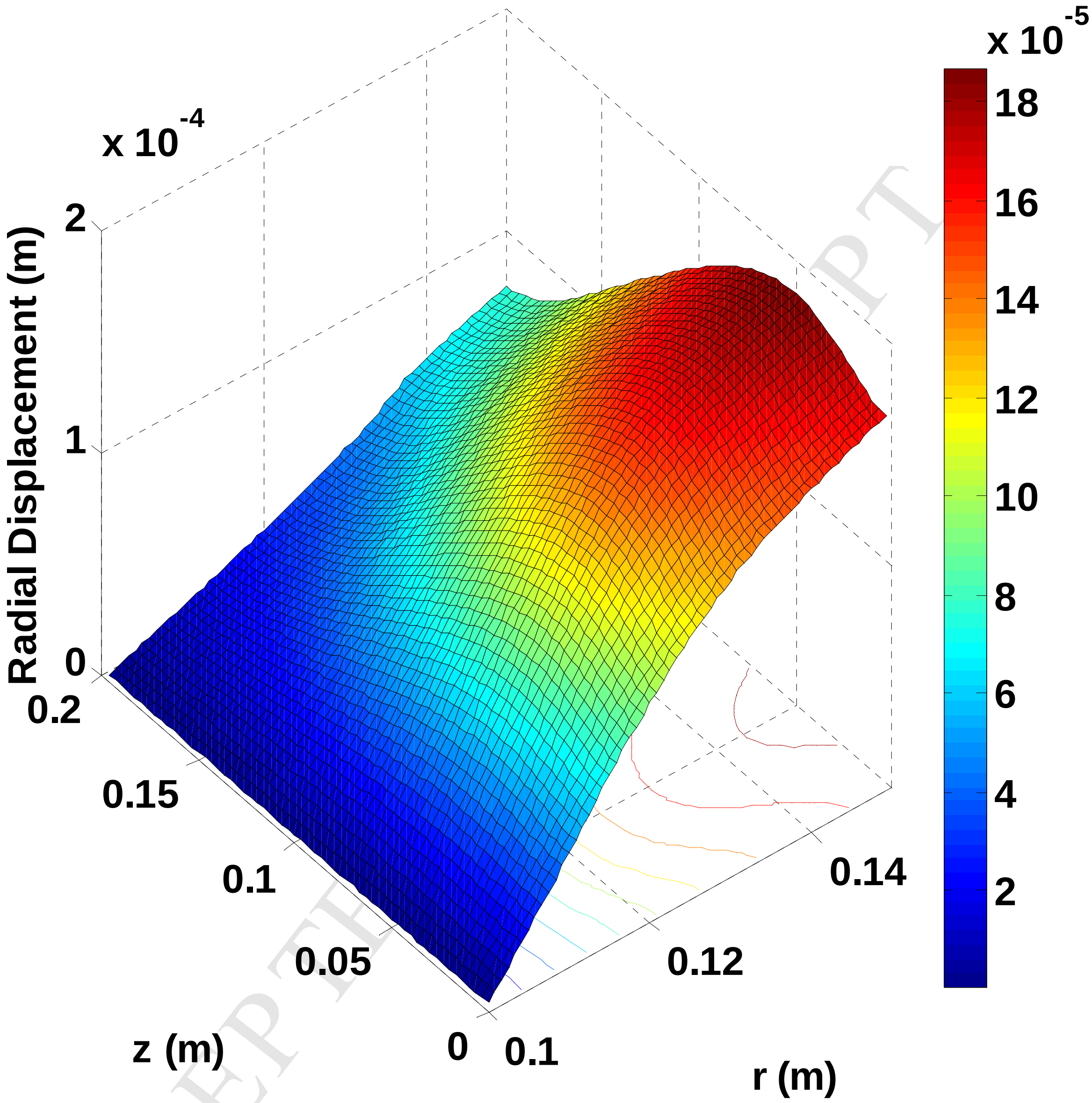

Figure 8: The contour of the radial displacement distribution through the cylinder wall for $n_r = n_z = 2$ at $t = 300$ seconds.

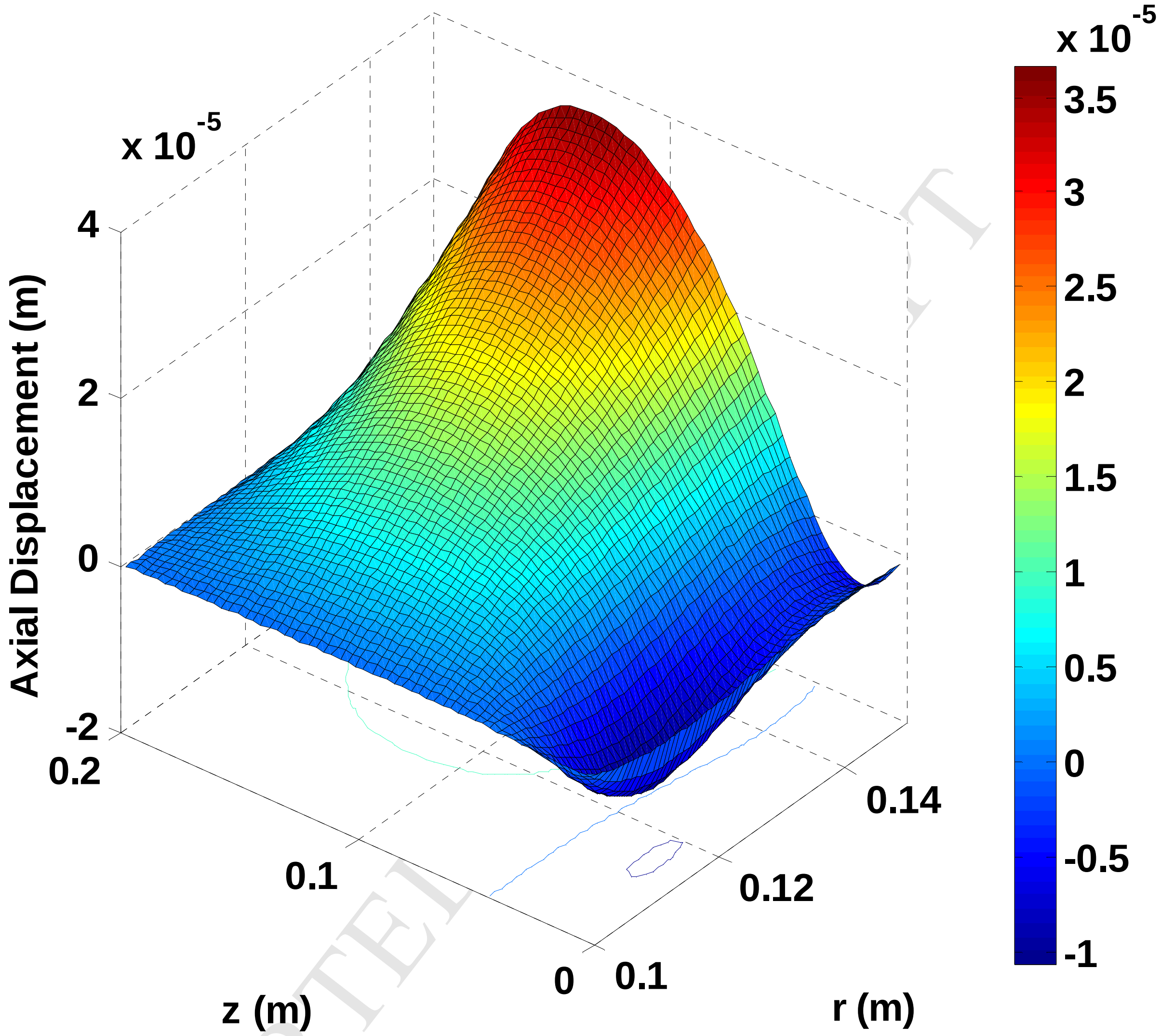

Figure 9: The contour of the axial displacement distribution through the cylinder wall for $n_r = n_z = 2 \; at \; t = 300$ seconds.

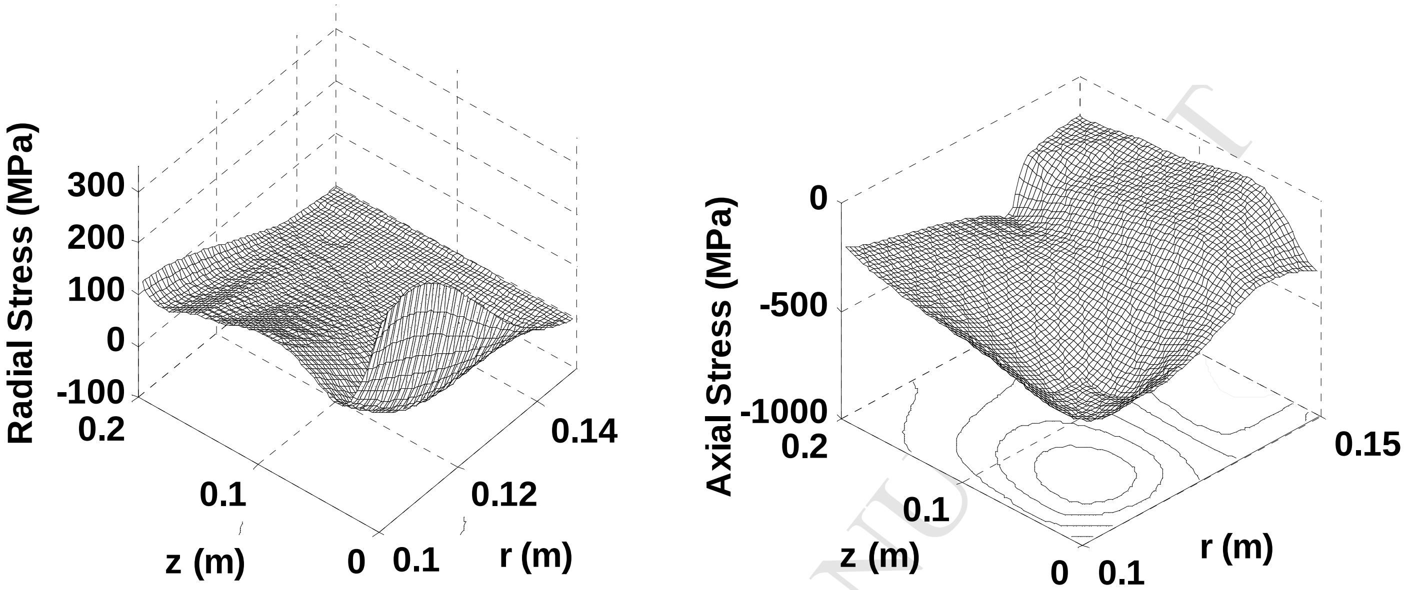

Figure 10: The radial and axial stresses distributions for $n_r = n_z = 2\ at\ t = 300$ seconds.

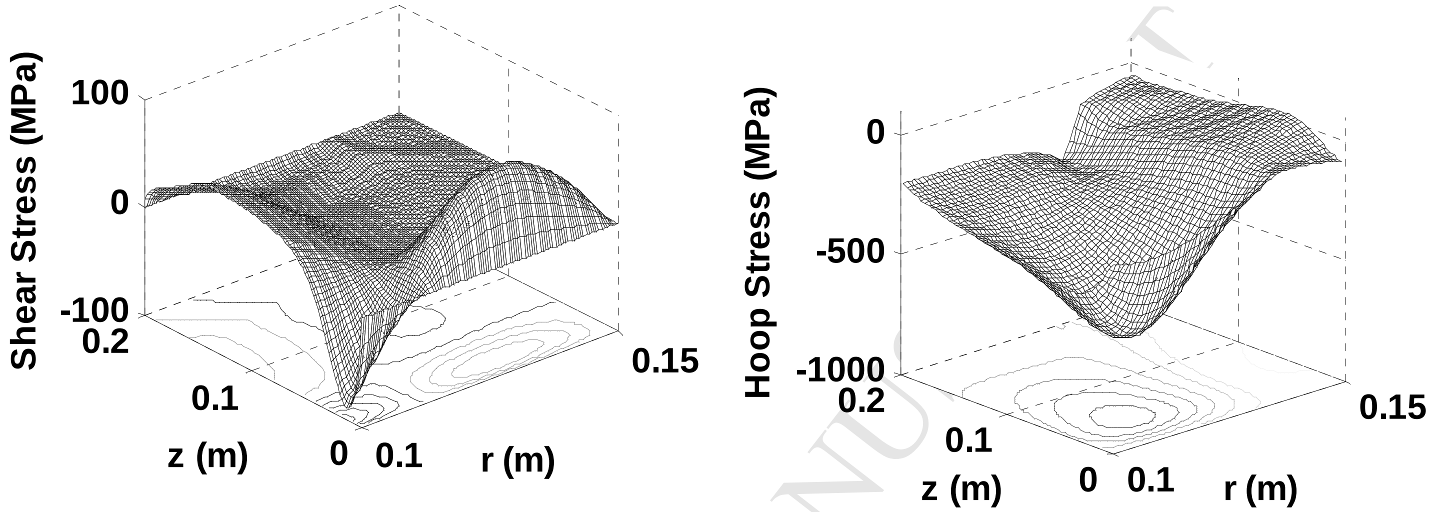

Figure 11: The shear and hoop stresses distributions for $n_r = n_z = 2\ at\ t =$ 300 seconds.

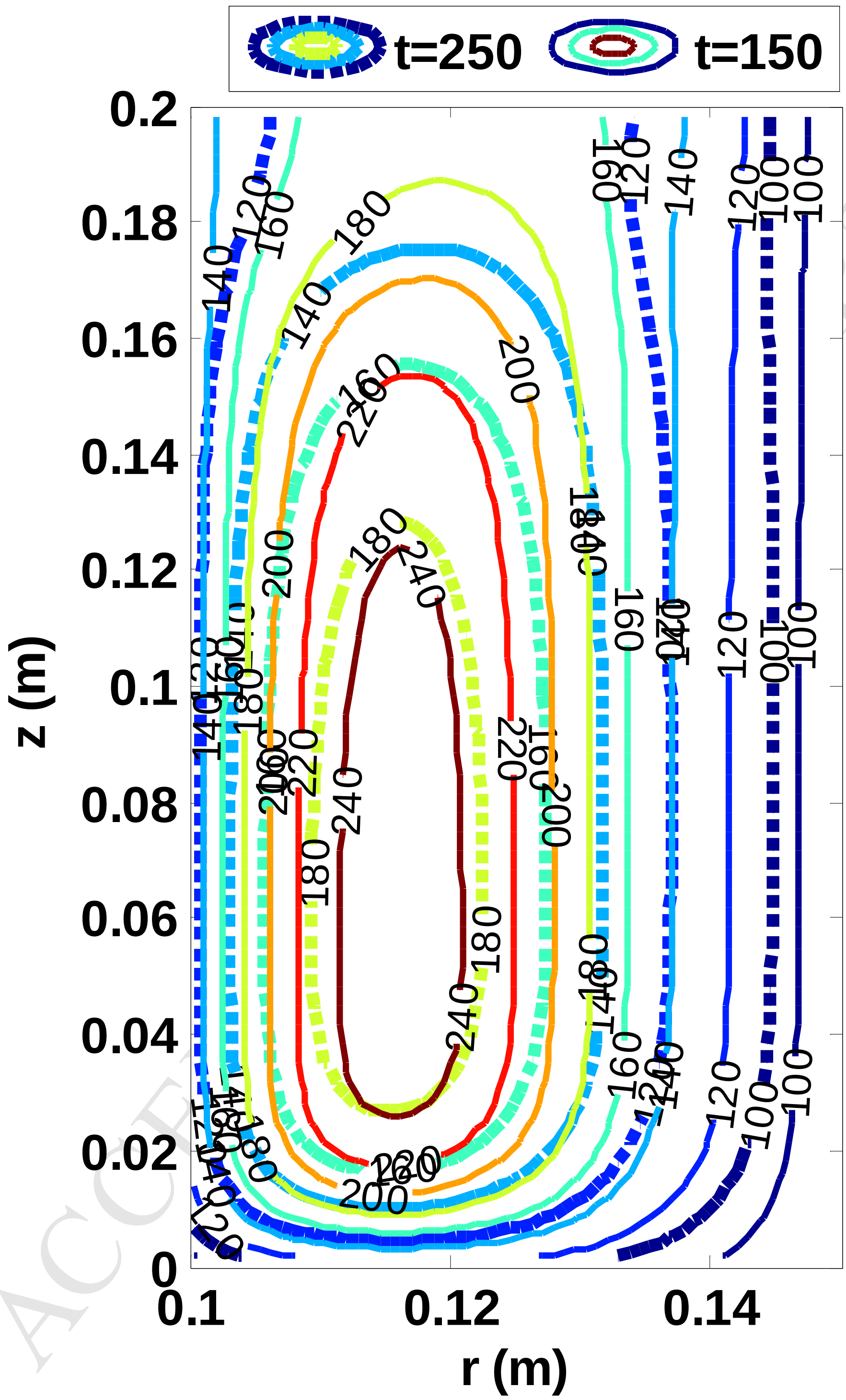

Figure 12: The contours of temperature distributions for two different times (t=150 and 250 s) for $n_r = 1\ and\ n_z = 5$.

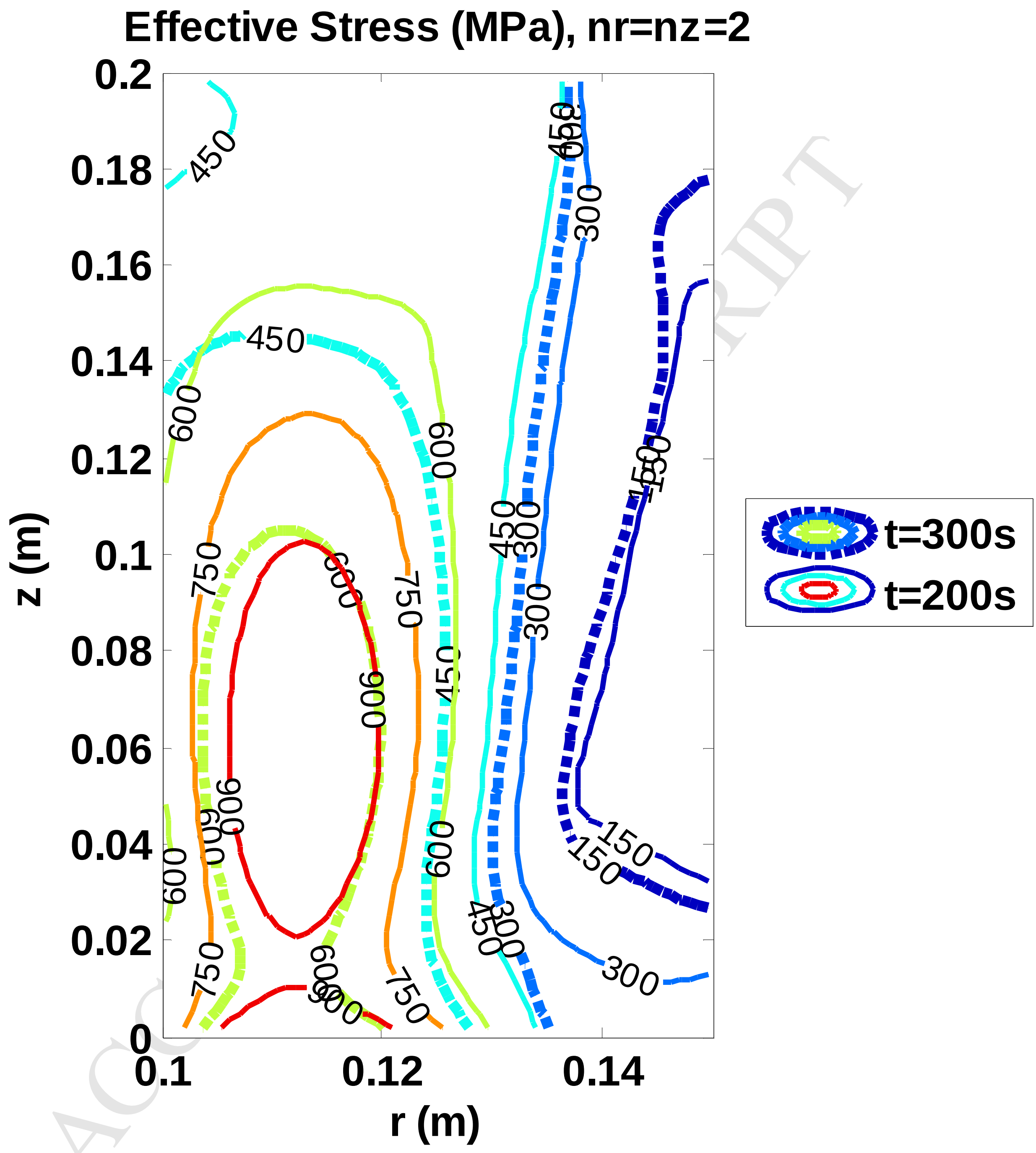

Figure 13: The contours of effective stresses through the cylinder wall for $n_r = n_z = 2$ and different time frames at 200 and 300 seconds.

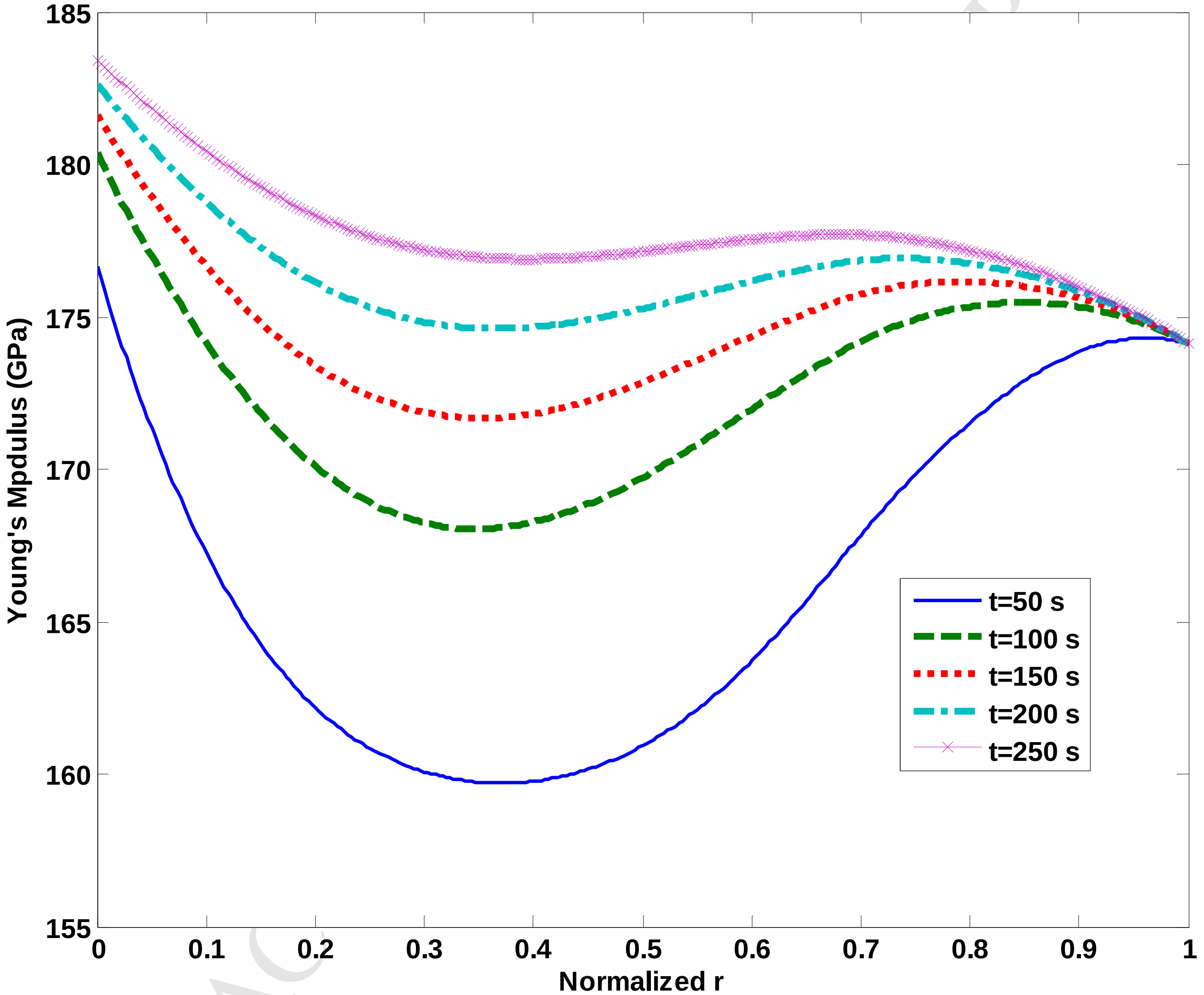

Figure 14: Effective Young's modulus along the horizontal centreline HCL versus normalized radial direction for $n_r = n_z = 2$ at different time frames.

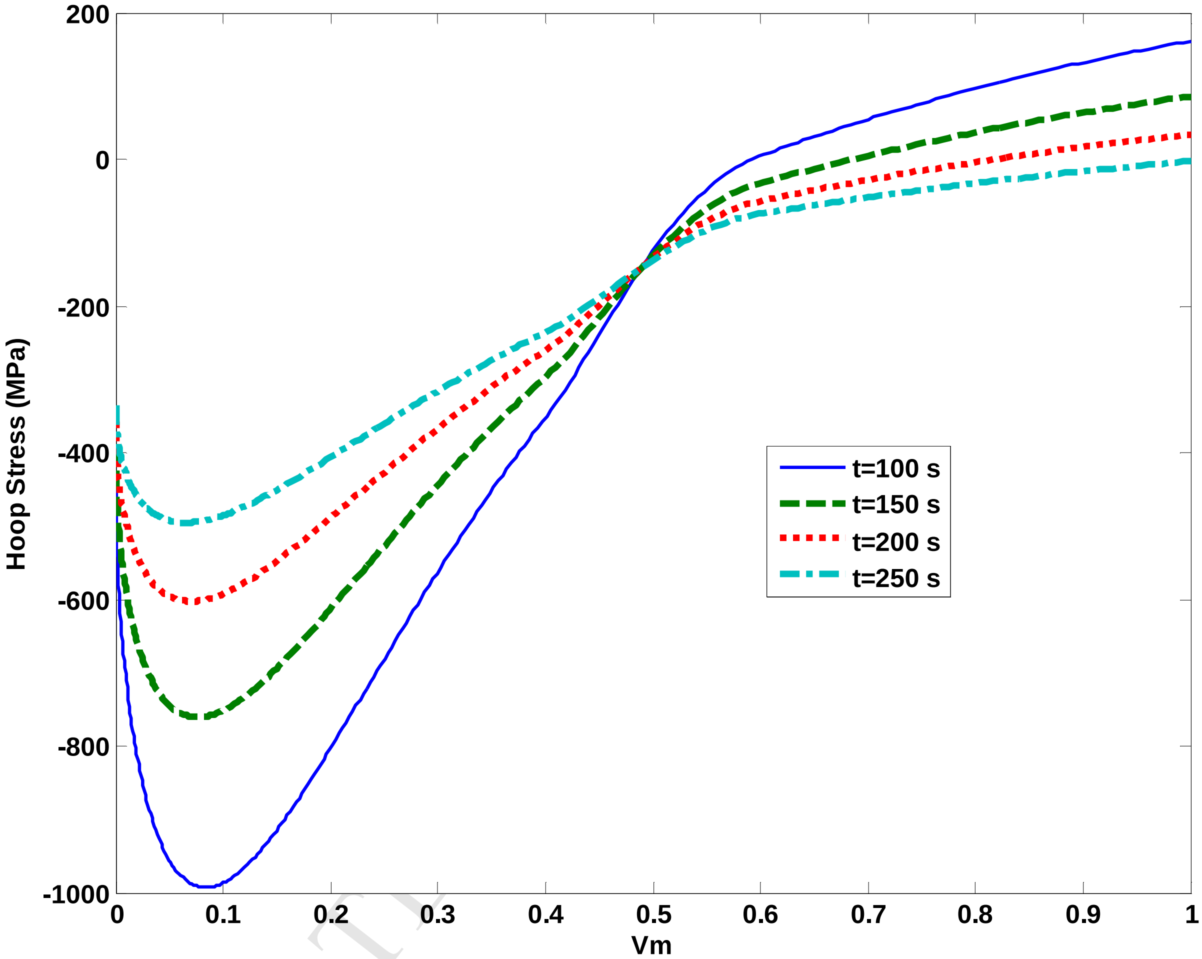

Figure 15: Transient hoop stresses according to the metallic volume fraction (Vm) along the HCL for $n_r = n_z = 2$and different time frames.

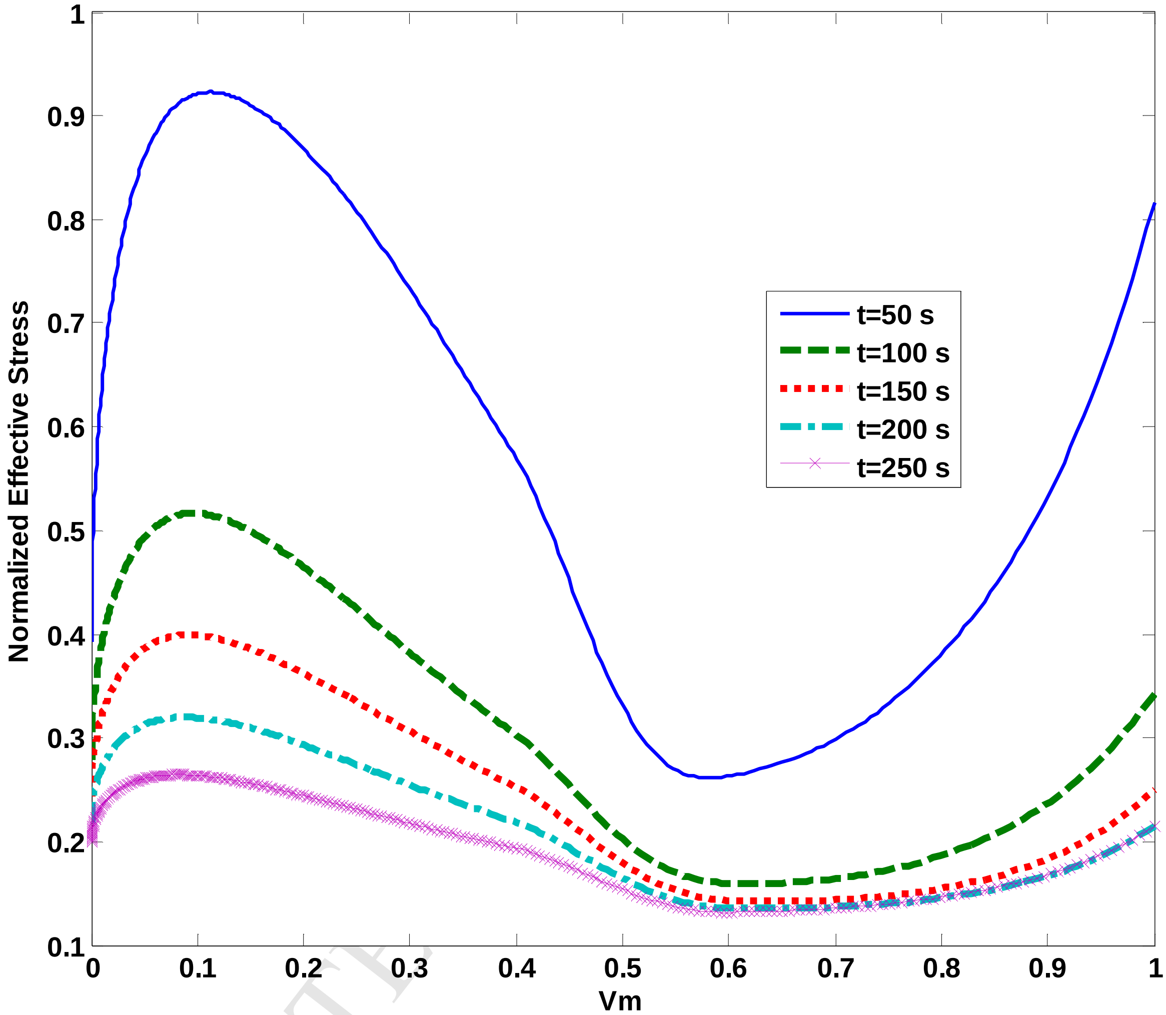

Figure 16: Transient normalized effective stress (NES) versus V_m along the HCL for $n_r = n_z = 2$ and different time frames.

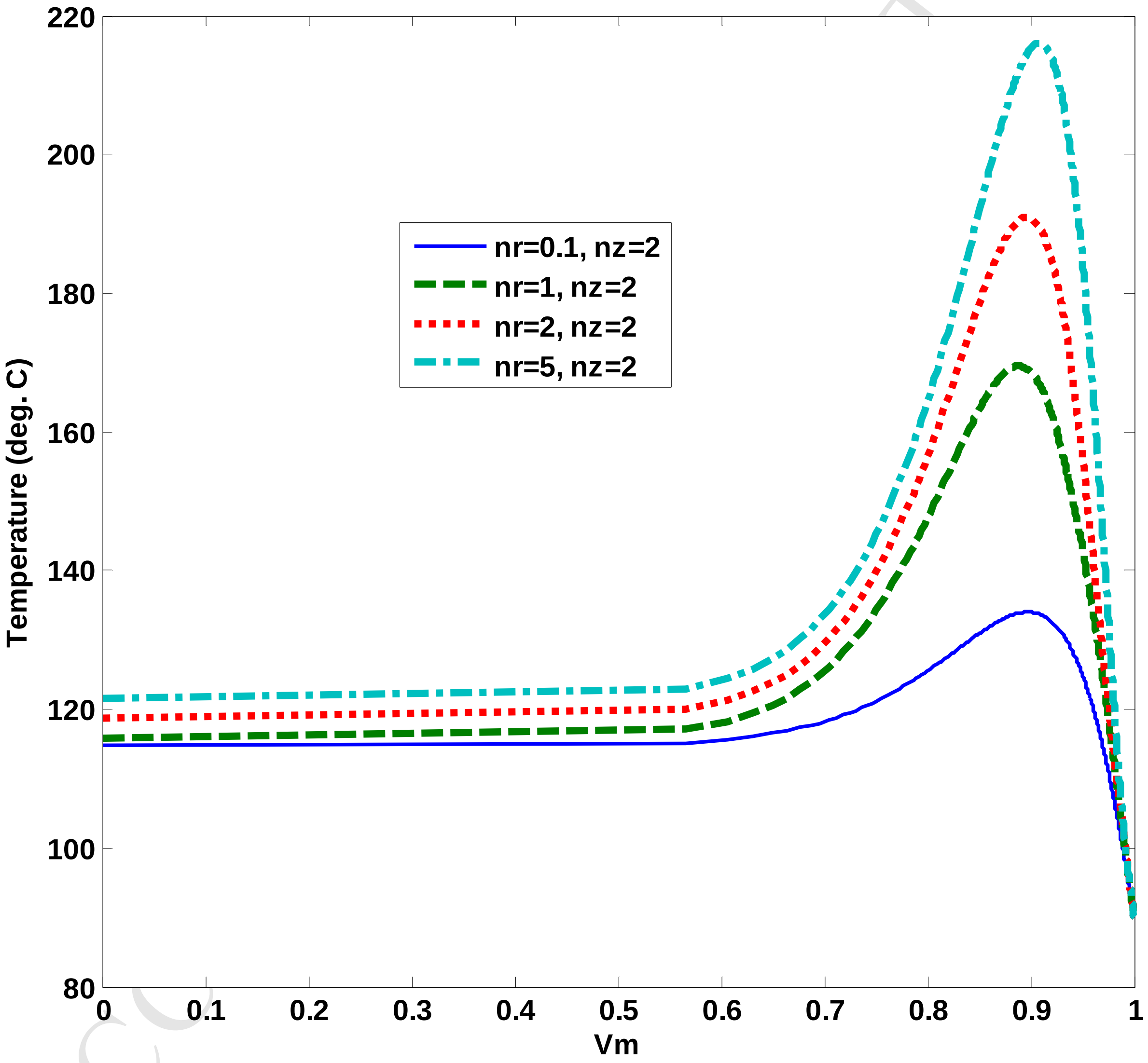

Figure 17:Temperature versus Vm along the HCL for $n_z = 2$ and different values of n_r at $t = 300$ seconds.

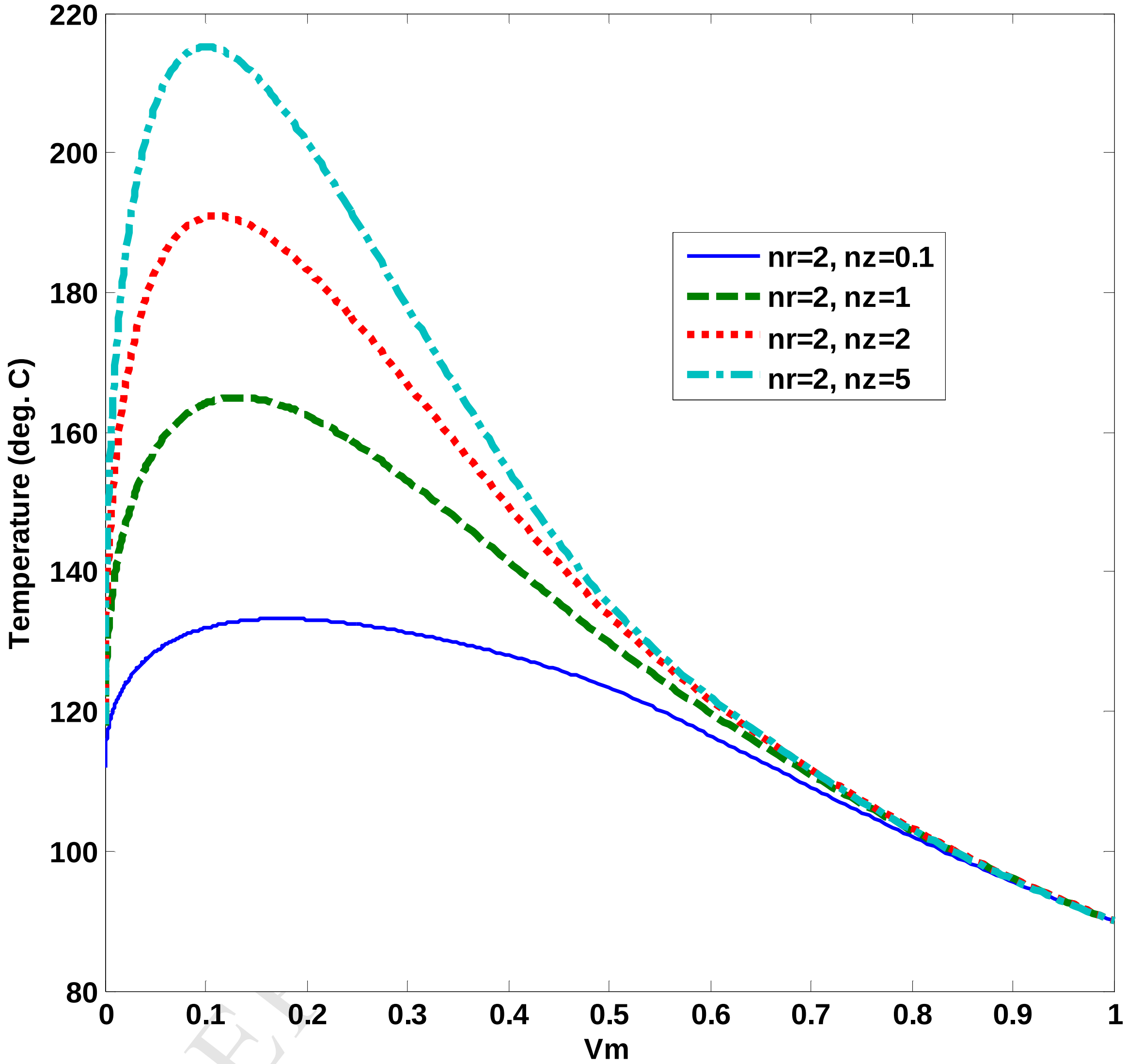

Figure 18: Temperature along HCL versus Vm for $n_r = 2$ and different values of n_z at $t = 300$ seconds.

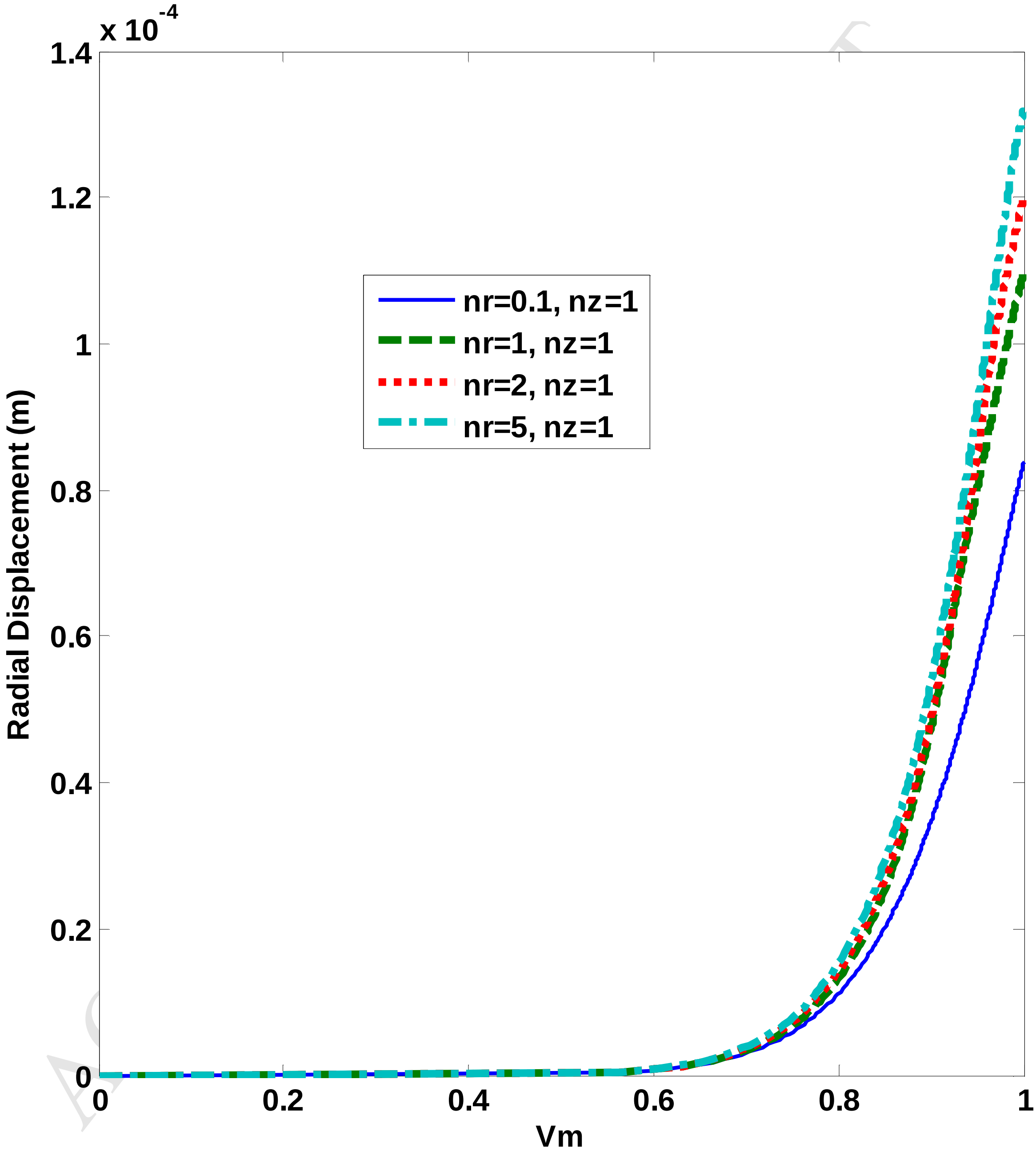

Figure 19: Radial displacement versus Vm along the HCL for $n_z = 1$ and different values of n_r at $t = 300\ s$.

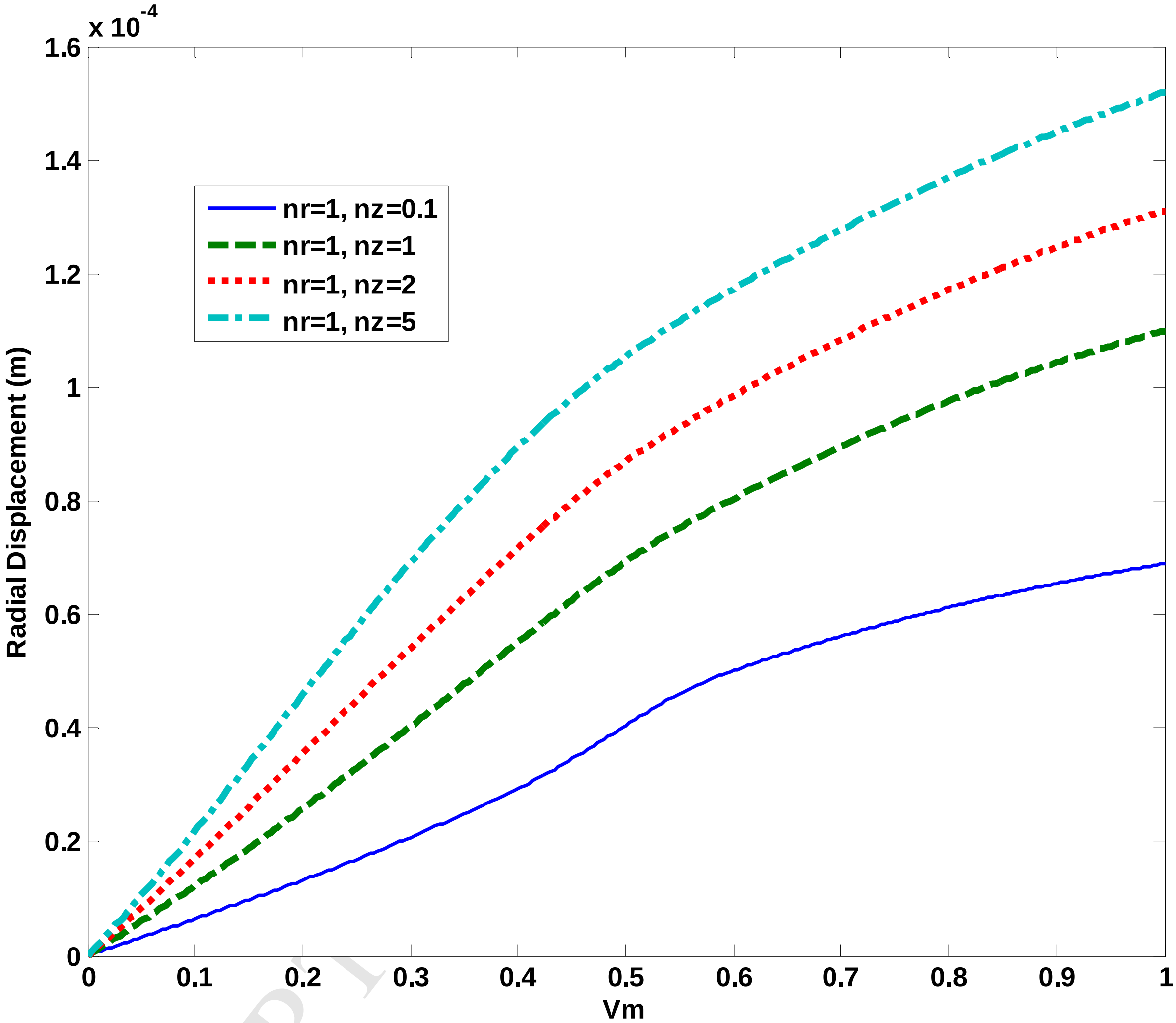

Figure 20: Radial displacement versus Vm along the HCL for $n_r = 1$ and different values of n_z $at\ t = 300\ s$.

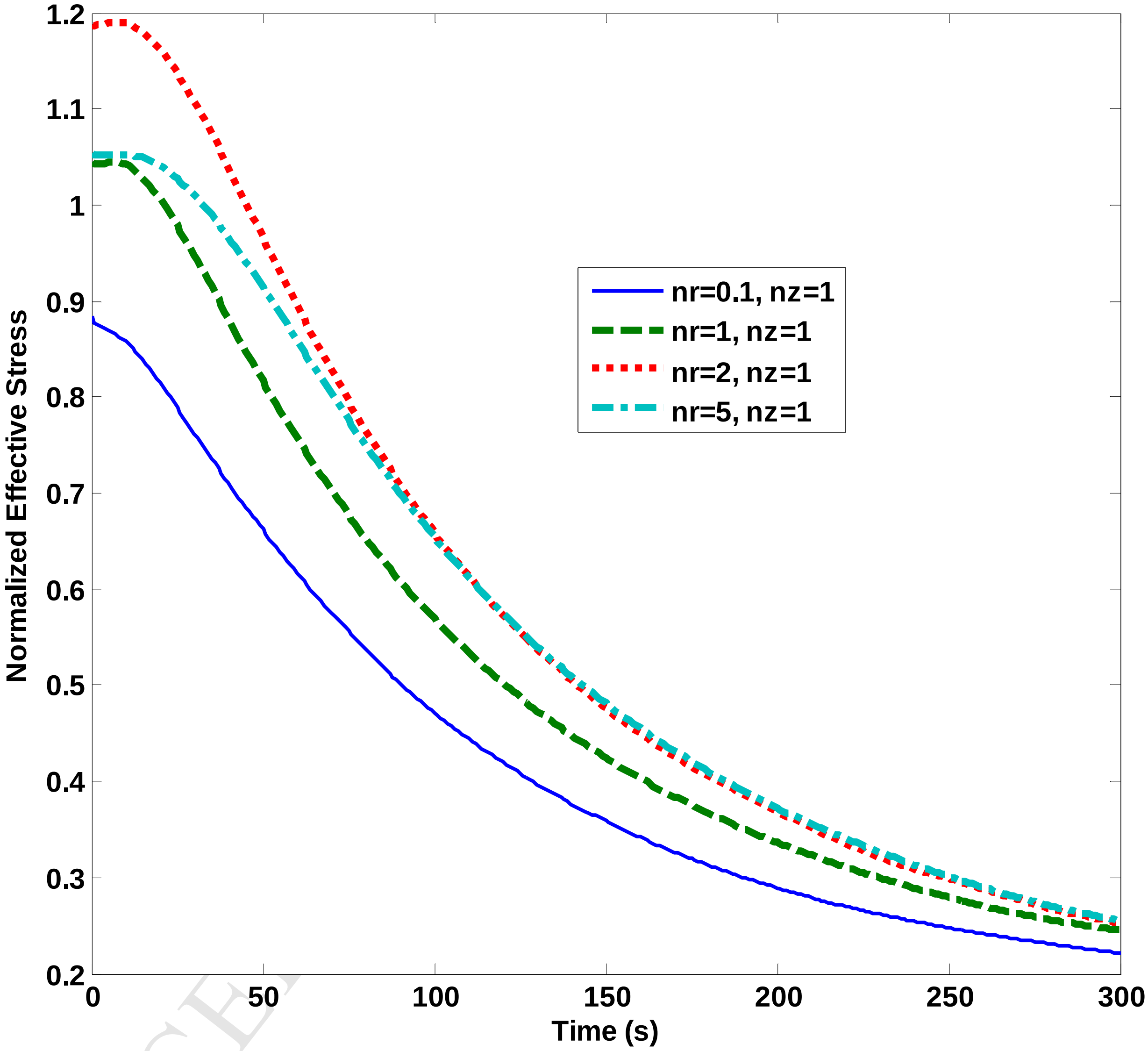

Figure 21: NES versus time at CP for different values of n_r and $n_z = 1$.

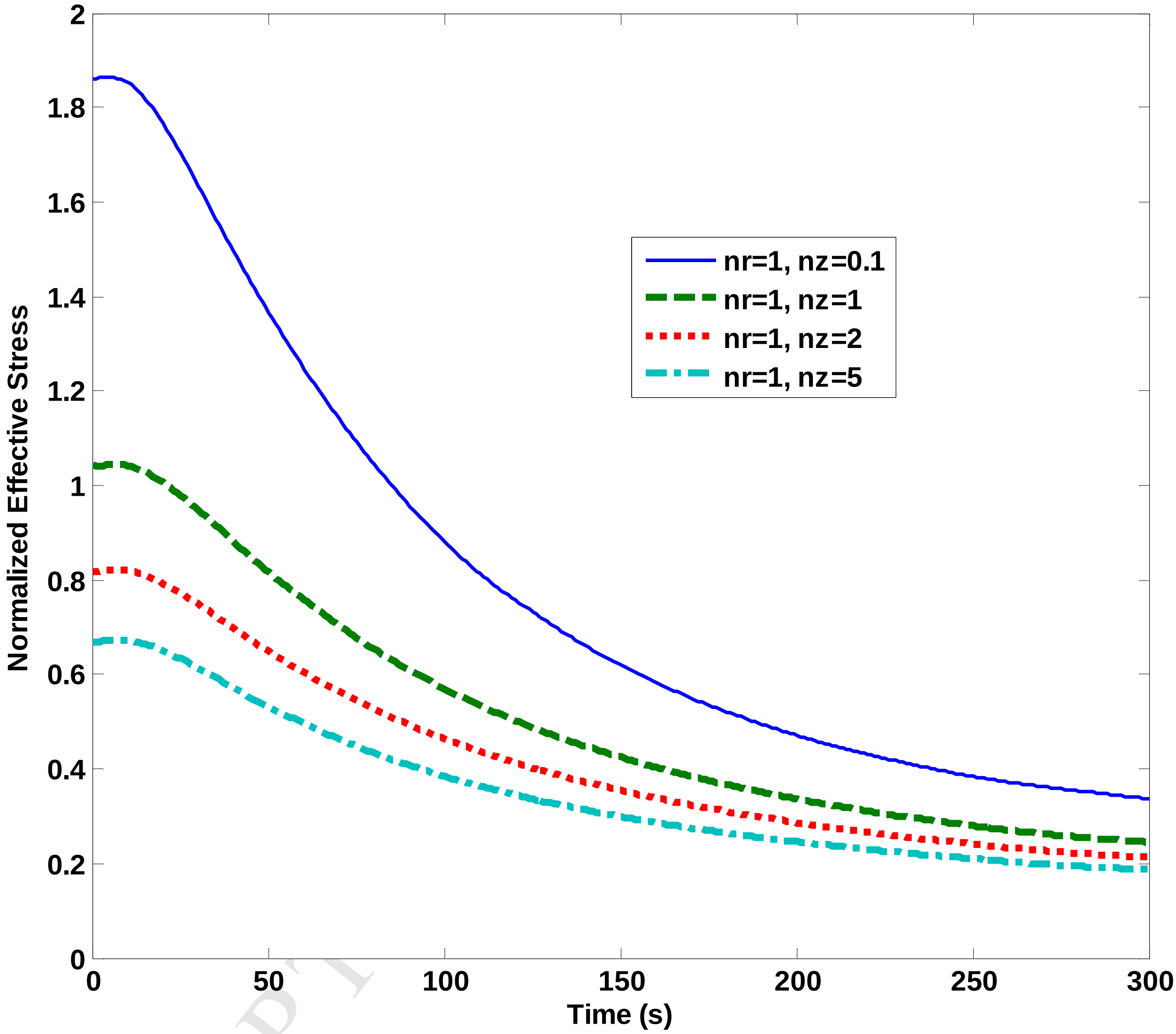

Figure 22: NES versus time at CP for different values of n_z *and* $n_r = 1$.

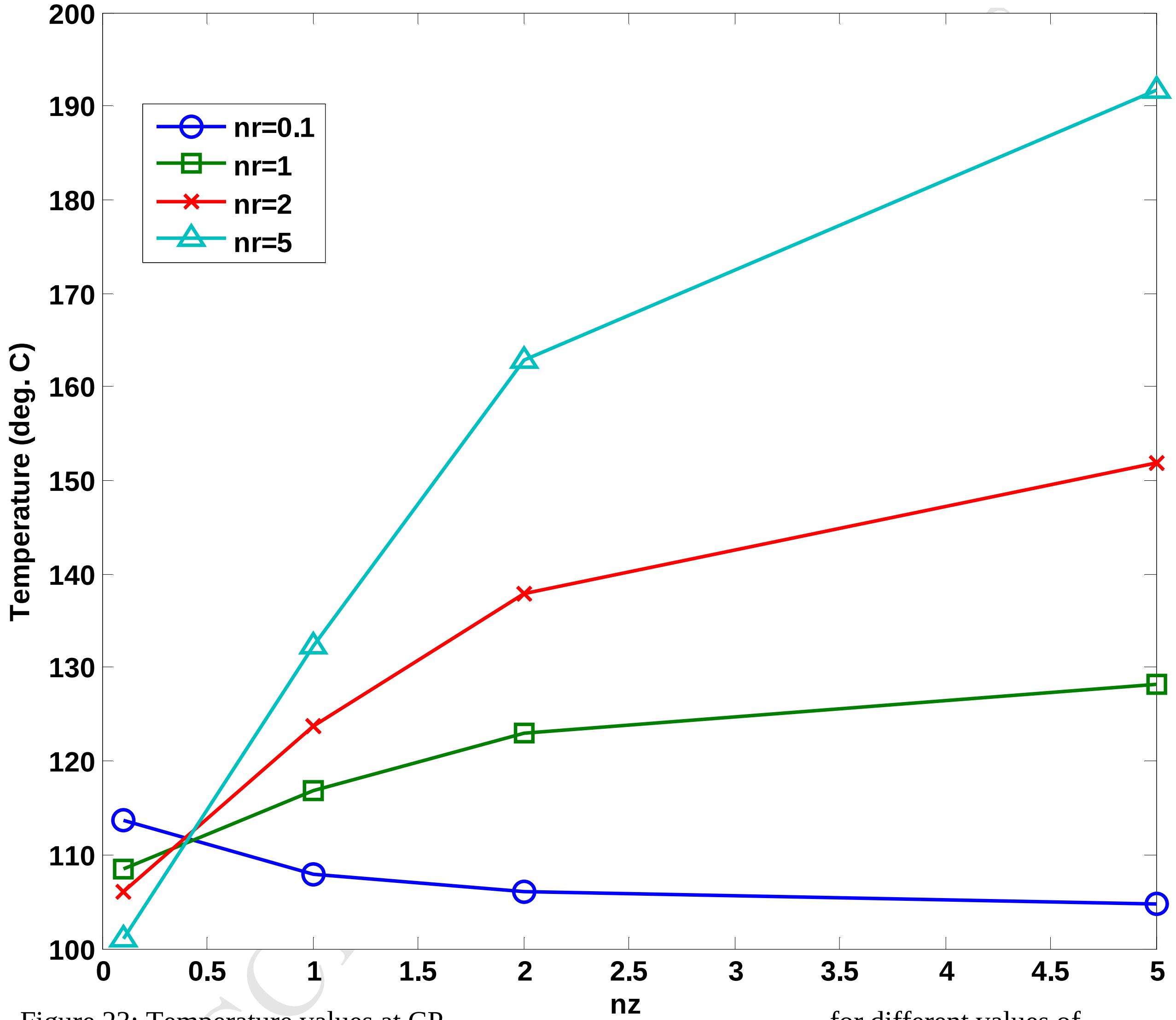

Figure 23: Temperature values at CP for different values of n_r and n_z.

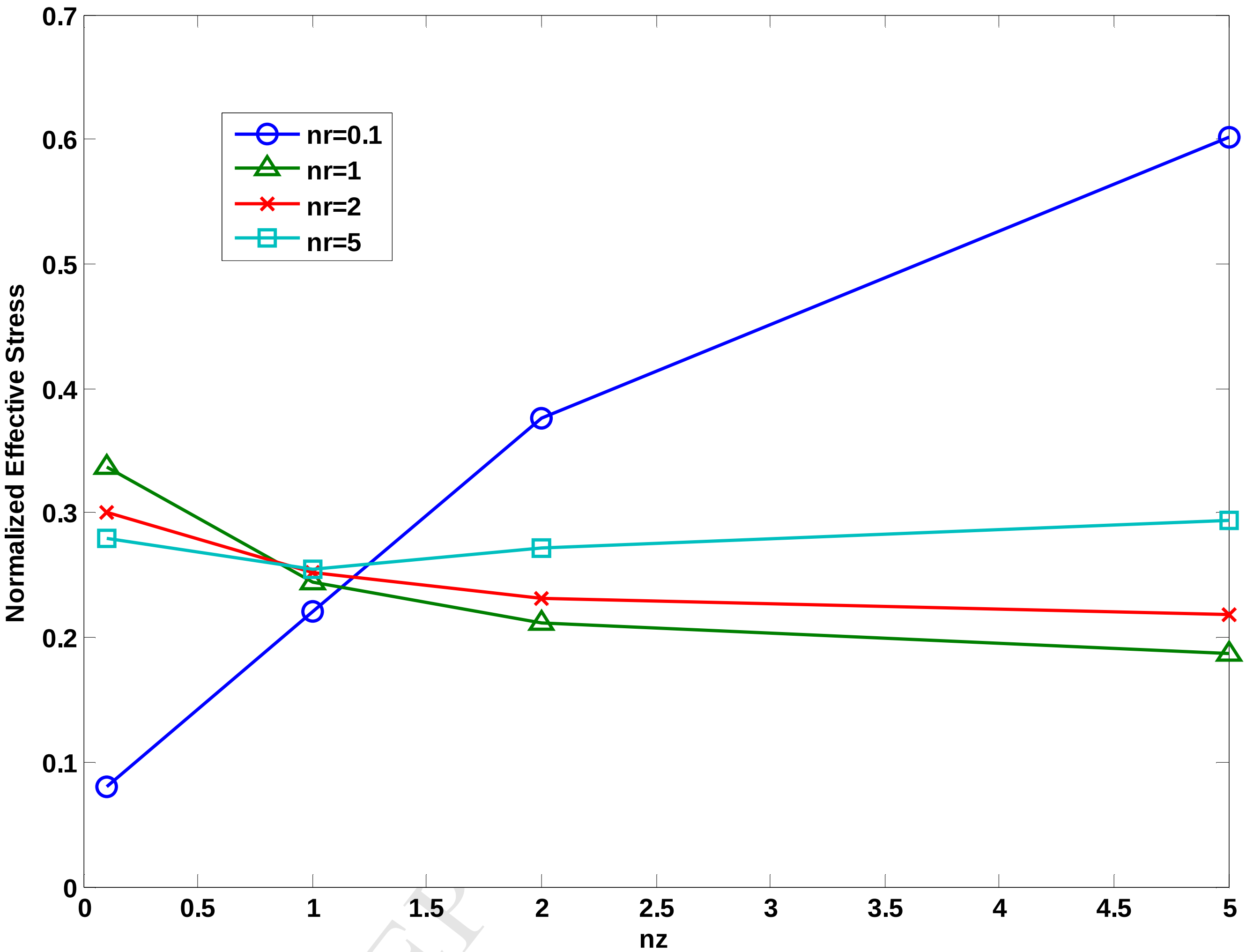

Figure 24: NES values at CP for different values of n_r and n_z.